PROJECT BOOK · FREDERICTON, NB, CANADA

THE FILAMENT FALCON

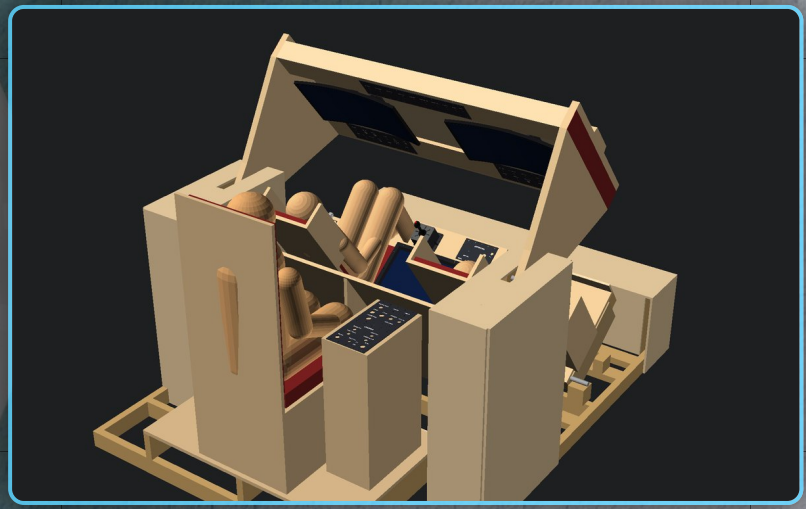
A home-built, three-seat spaceship cockpit simulator for a dad and his two kids: reclining launch seats, a drop-down instrument pod, 3D-printed backlit panels, three ESP32-S3 joysticks and a Godot space game with six missions.

PILOT · left seat | OPS · right seat | MISSION CONTROL · Dad

Yves Daigle · github.com/whyvas/filament-falcon

Book v1.0 · generated 2026-10-12 09:44 ADT (Atlantic time) · main @ 5858783 + local changes

Status: v0.1 design and software. Nothing has been physically printed, wired or flashed yet.



CAD render: flight configuration · background: docking (in-game)

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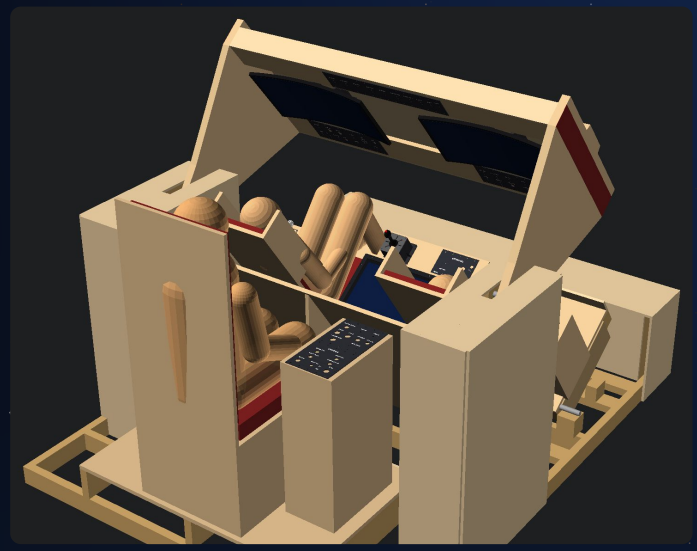
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WHY WE ARE BUILDING A SPACESHIP AT HOME

01 PROJECT VISION

A home-built, three-seat spaceship cockpit simulator for a dad and two kids. The two kids fly as **Pilot** (left) and **Ops** (right), and Dad runs **Mission Control** from the seat behind them. The seats recline back, then a pod with the window monitors and instrument panels lowers into place in front of the crew. Every panel is 3D printed with backlit white lettering lit by a single WS2812B strip.

This book collects everything designed so far into one place: how the cockpit moves, how it is kept safe, how the wooden frame and printed panels go together, how the electronics talk to the PC, and what the game asks of each crew member. It is generated straight from the project files, so it can be rebuilt whenever the design changes.



Flight configuration. Kids reclined, pod lowered with both 27" window monitors, dash and overhead panels; Dad at Mission Control behind.

3
seats: two kids side by side in front, Dad behind on a 250 mm (9.8") dais

1960×2450 mm
footprint (6' 5.2" × 8' 0.5")

35°
seat recline, astronaut launch-style, then the pod drops down

6
missions in the Godot game, each with a hero moment for both kids

11
3D-printed backlit panels with 70 through-letter labels

242
addressable LEDs on ONE WS2812B data line (placeholder map)

3
ESP32-S3 boards, each a real USB joystick: Falcon Pilot, Ops, Command

$3,447
CAD still to buy (before tax), from the BOM spreadsheet

CREW ROLES

three seats, three different jobs

PILOT · LEFT SEAT

PILOT (left seat) flies the Falcon with the big stick and throttle. You point the ship, fire the engine, land it, and fire the mining laser.

rotation and main engine. Joystick = pitch/roll/yaw + trigger. Throttle lever. Engine arm (covered toggle), ignition button, landing gear lever, flight-assist mode rotary (OFF / STAB / PROGRADE / HOLD), airbrake/flaps, boost, view cycle, horn.

USB device `Falcon Pilot` · 4 axes · 15 mapped buttons

OPS · RIGHT SEAT

OPS (right seat) runs the ship. You handle power, shields, the RCS stick that slides the ship sideways, the robot arm and tractor beam, the scanner, the jump drive, and fixing things when they break.

translation, systems, and payload. Joystick = RCS translate X/Y + Z(twist) + trigger (also drives the robotic arm, tractor beam, and sensor camera depending on mode). Power distribution toggles (REACTOR, LIFE SUPPORT, SHIELDS, COMMS, SENSORS), mode rotary (RCS / ARM / SCANNER), cargo bay doors, docking clamp, scanner ping, fire suppression, nav target select, jump/warp arm + engage (covered).

USB device `Falcon Ops` · 3 axes · 17 mapped buttons

MISSION CONTROL · DAD

COMMANDER (a grown-up, or whoever is not flying) sits at the Commander screen: picks the mission, checks the harnesses, presses START, watches both kids' tasks on MISSION CONTROL, and can make things go wrong on purpose (engine fire, cabin leak, power fault, sensor glitch) to see if the crew can fix it.

crew-motion hold switch + pod release lever (hardwired), key switch MASTER POWER, mission select rotary, START / PAUSE, MASTER CAUTION reset, failure-injection buttons (ENGINE FIRE, CABIN LEAK, POWER FAULT, SENSOR GLITCH), INTERCOM push-to-talk and ALL-CALL/PA buttons, cockpit lights dimmer.

USB device `Falcon Command` · 1 axes · 27 mapped buttons

SHARED OVERHEAD CONSOLE

launch checklist switch row (BATTERY, FUEL PUMPS, AVIONICS, APU START, GUIDANCE, IGNITION ARM), cabin lights, beacon, strobe, EJECT (covered, just for fun), each with a backlit label and status LED.

KIDS' CONSOLE TOUCHSCREEN

attitude ball, speed/altitude tapes, fuel, engine/gear/flaps/assist, power distribution toggles, RCS/ARM/SCANNER mode, cargo doors, docking clamp, nav target and steer-to, overhead checklist status, camera feeds (docking/arm/chase), each kid's current task and MASTER CAUTION, a TALK pad per kid. Every touch control mirrors a hardware control; plugged-in hardware wins. While a board is unplugged its panels are drawn on screen (virtual panels, generated from the CAD layouts; guarded switches need two taps), with a touch joystick for a missing stick; the same for Falcon Command on the Commander screen.

COMMANDER SCREEN

mission select and briefing, START/PAUSE/RESTART/skip step, harness check, failure injection, MASTER CAUTION reset, difficulty and assist settings, time warp, live score, a mission-control overview (orbit map, telemetry, crew status with both kids' current tasks, ship health, faults, crew position, intercom TALK / ALL-CALL / mutes), controls test, board status (plugged in or virtual), the Falcon Command panels on screen, the intercom page, and SETUP (screen roles, headsets, quality, FPS cap, vsync).

WHY BOTH KIDS GET A HERO MOMENT

from the game's crew guide

MISSION	PILOT HERO MOMENT	OPS HERO MOMENT
First Light (launch) Countdown, lift-off, fly to orbit and drop off a satellite.	Lighting the engine at T−5 and steering into orbit.	Staging the booster and releasing the satellite with the arm.
Space Hug (docking) Glide into the space station slower than a snail.	Keeping the nose dead-on with HOLD.	Creeping in with the RCS stick on the docking camera, then DOCK CLAMP!
Eagle Eye (lunar) Skim the Moon, find the beacon and land like a feather.	Slowing down and hovering to a soft touchdown.	Finding the beacon with the scanner and sliding the ship over it.
Rock Hoppers (asteroids) Rescue a lost pod, mine three crystals, jump home.	Dodging rocks and lasering crystal rocks.	Tractor-beaming the pod and crystals aboard, then the JUMP!
Homecoming (reentry) Blaze through the fiery air and glide onto the runway.	Holding the nose up, then flying the green boxes down to the runway.	Shields, beating the radio blackout, finding the runway.
Free Flight (sandbox) No rules: visit every place with the jump drive.	Try ASSIST OFF!	Play with the arm, scanner and jump drive.

WHY THE JOBS ARE BALANCED

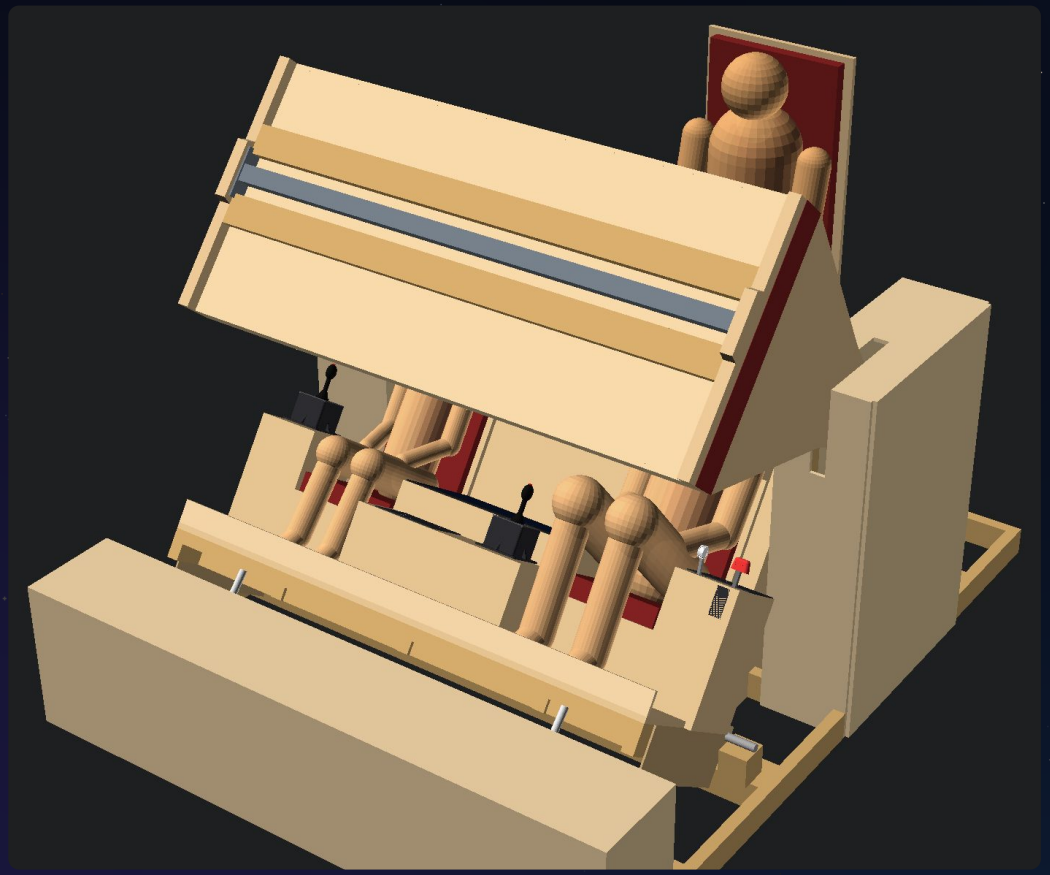
Each mission gives both seats a hero moment (e.g. Pilot flies the launch and landing; Ops does the docking translation, arm capture, and hyperspace jump).

The two kids never fight over one stick: the Pilot owns **rotation and the main engine**, Ops owns **translation, systems and payload**. Most missions need both: docking needs the Pilot to hold the nose while Ops creeps in on RCS; the asteroid rescue needs the Pilot to laser the rocks while Ops tractor-beams the crystals and fires the jump drive.

Dad is deliberately **not a third player**: he runs the checklist, starts missions, injects failures (engine fire, cabin leak, power fault, sensor glitch) for the crew to fix, and is the safety observer who alone can move the seats and the pod.

TWO KIDS IN FRONT, DAD BEHIND

02 COCKPIT LAYOUT AND DIMENSIONS

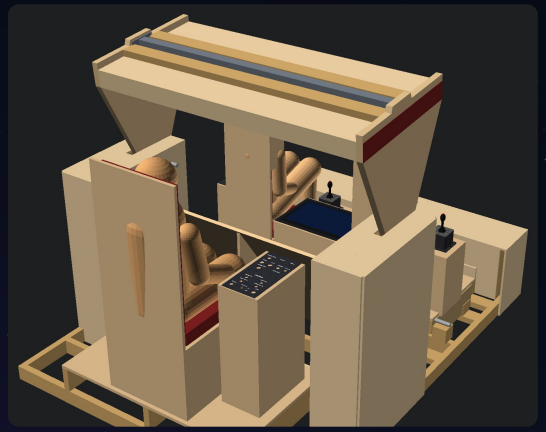


Flight, from the front. The pod (two monitors, four dash panels, three overhead panels) is down; the TD2220 touchscreen rides in the centre console between the kids.

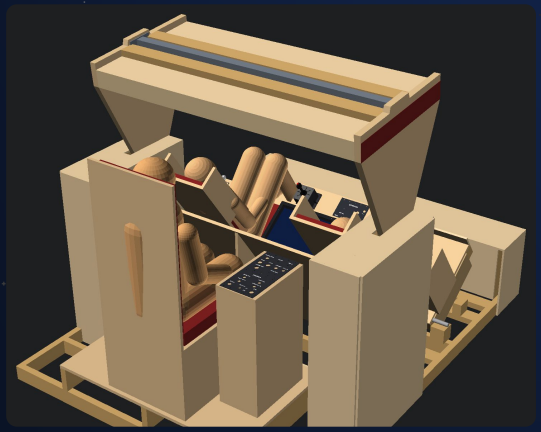
DIMENSION	VALUE
Footprint (W × L)	1 960 mm (6' 5.2") × 2 450 mm (8' 0.5")
Casing / tower height	1 112 mm (3' 7.8")
Swept height (pod parked up)	1 831 mm (6' 0.1") → ceiling ≥ 1 950 mm (6' 4.8")
Pod lowest point when parked	1 639 mm (5' 4.5") above the floor
Kid seat centres	x = ±460 mm, 920 mm (36.2") apart, so the 559 mm (22") TD2220 trim fits between
Cradle pivot	840 mm (2' 9.1") behind the nose, 200 mm (7.9") up
Pod pivot (reclined head centre)	1 375 mm (4' 6.1") behind the nose, 1 005 mm (3' 3.6") up
Screens from pod pivot	600 mm (1' 11.6")
Eye → screen, Pilot (12 yr)	542 mm (1' 9.3"), gaze 40° up
Eye → screen, Ops (6 yr)	590 mm (1' 11.2"), gaze 55° up (60-80 mm booster cushion recommended)
Dad's dais	250 mm (9.8") high; fixed seat, no recline, no pod

Numbers read from `frame/generated/geometry.json` (written by `frame/frame_model.py`).

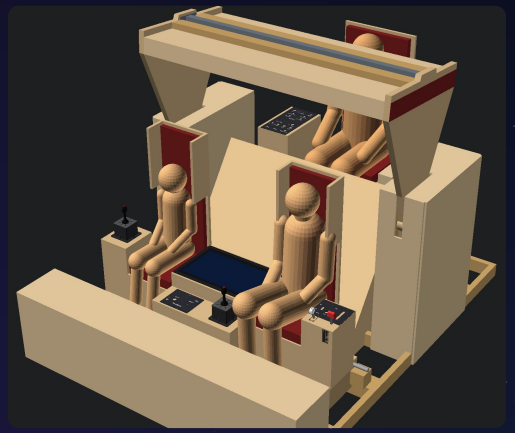
THE COCKPIT IN ITS STATES



1 · Boarding. Seats upright, pod parked up over the kids.



2 · Reclined. The cradle has tipped back 35°; pod still locked up.

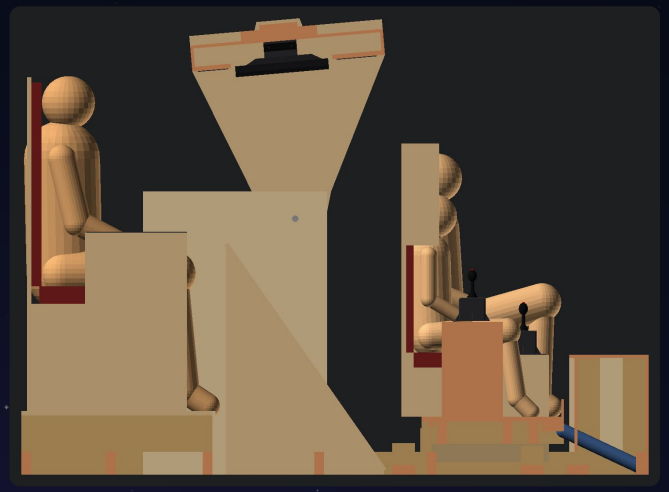


Boarding, from the front. Side entry gaps in front of the casings.

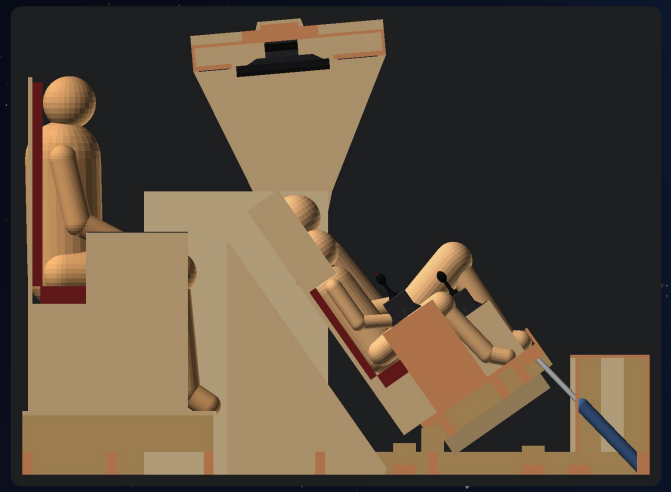


Dad's view. From Mission Control he sees both kids, their hands and the pod gap, plus the front monitors.

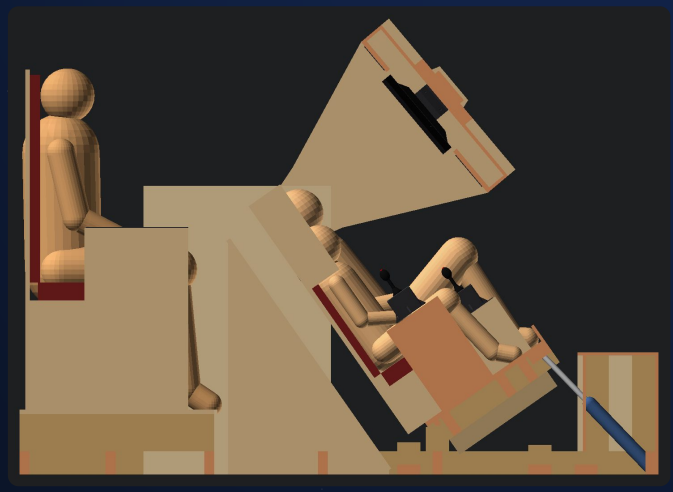
SIDE SECTIONS AND KEY NUMBERS



Side section · boarding



Side section · reclined



Side section · flight

KEY NUMBERS (FRAME README)

- Footprint 1960 W x 2450 L mm (6' 5" x 8' 0.5"); casings/towers 1112 mm high; **swept height 1828 mm (pod parked up) → ceiling ≥ 1950 mm**.
- Seat centres x = ±460 mm (920 apart) so the 559 mm TD2220 trim fits between them.

CLEARANCES AND SIGHT LINES

- Clearances (worst case = 12 yr manikin, pod at every angle): pod chin to thighs 195 mm, shins 220, head ≥ 251, upper arm to cheek 157; kid arm to casing 35 (finger-safe 25+); seat back to casing 60; cradle to nose wall ≥ 30; cheek to inner wall / post 66 / 50.
- Eye → screen centre: 12 yr 542 mm (gaze 40° up from horizontal), 6 yr 590 mm (55°) → a 60-80 mm booster cushion for the 6 yr is recommended.

RECLINE, THEN DROP THE POD

03 MOTION SYSTEM



The flight sequence. 1 boarding (seats upright, pod parked up) → 2 recline (cradle tips back 35°) → 3 flight (pod swings down in front of the crew). Exit is the exact reverse: pod up first, then seats upright.

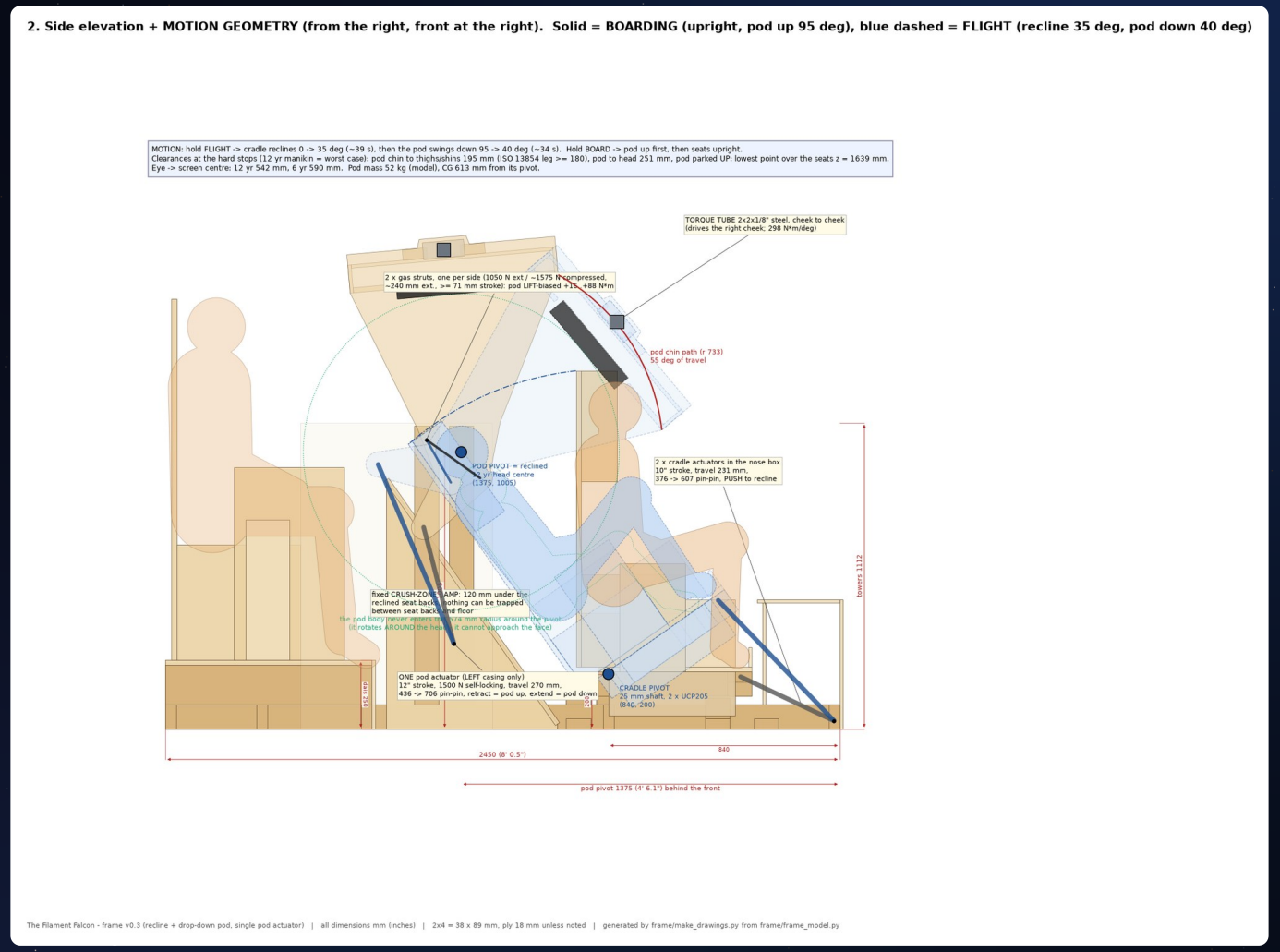
STEP	WHAT MOVES	DRIVE	END STOPS / LOCKS
1 BOARDING	nothing: seats upright, pod parked UP over the kids (lowest point 1639 mm over the floor)	-	pod locked UP by spring pawls
2 RECLINE (hold FLT)	the **cradle** (both kid seats, side consoles, sticks, throttle, gear lever, TD2220 centre console) tips back 35° about a 25 mm shaft at 840 mm behind the nose, 200 mm up	2 × 12 V actuators, 10" (254 mm) stroke, in the nose box, PUSH to recline	rubber stop blocks on a steel-angle stop rail at 35°; rests on the front pads at 0°
3 POD DOWN (keep holding FLT + release lever)	the **pod** (both monitors, 4 dash panels, 3 overhead panels) swings 55° down (95 → 40°) about a 25 mm shaft at the reclined 12 yr's head centre (1375 mm behind the nose, 1005 mm up)	**ONE** 12 V actuator, 12" stroke, 1500 N self-locking, in the LEFT casing (extends to lower, retracts to raise); a steel torque tube carries the drive to the right cheek; 2 gas struts (one per side) make the pod slightly LIFT-biased	6 mm steel stop pins on each cheek; pawls drop into the DOWN notch
EXIT (hold BRD)	pod up first, then cradle upright	same	same

Each motion takes ~35-40 s (slow on purpose, ≤ 10 mm/s at the actuator).

GEOMETRY: WHY THE POD PIVOTS ON THE HEAD

Why the pod pivots on the head: the pod pivot is the reclined 12 yr's head centre, so every pod point moves on a circle around the head and can never come closer to the face than at its parked positions. The screens stay 600 mm from the pivot (~540 mm from the 12 yr's eyes, ~590 mm from the 6 yr's).

Dad's seat does not recline and has no pod. He is the safety observer: the motion controls are only at his seat, hold-to-run, so he must see both kids' heads, hands and the pod gap the whole time it moves. A reclining dad seat would put his eyes below the pod's back and hide the kids; a third pod would need its own actuators, interlock and clearance envelope next to the kids' pod (pinch point between two moving bodies) for little gain. He keeps the CMD console (2 printed panels) and gets a view over the kids at the front monitors (`overview_dad_view.png`); add a small fixed 3rd display at the CMD console later if wanted (needs a GPU output, see `docs/mechanical/displays.md`).



Drawing 02 · side view with both motions: cradle and pod pivots, arcs, actuators, gas struts, eye-to-screen lines and clearances.

FORCES, ACTUATORS AND GAS STRUTS

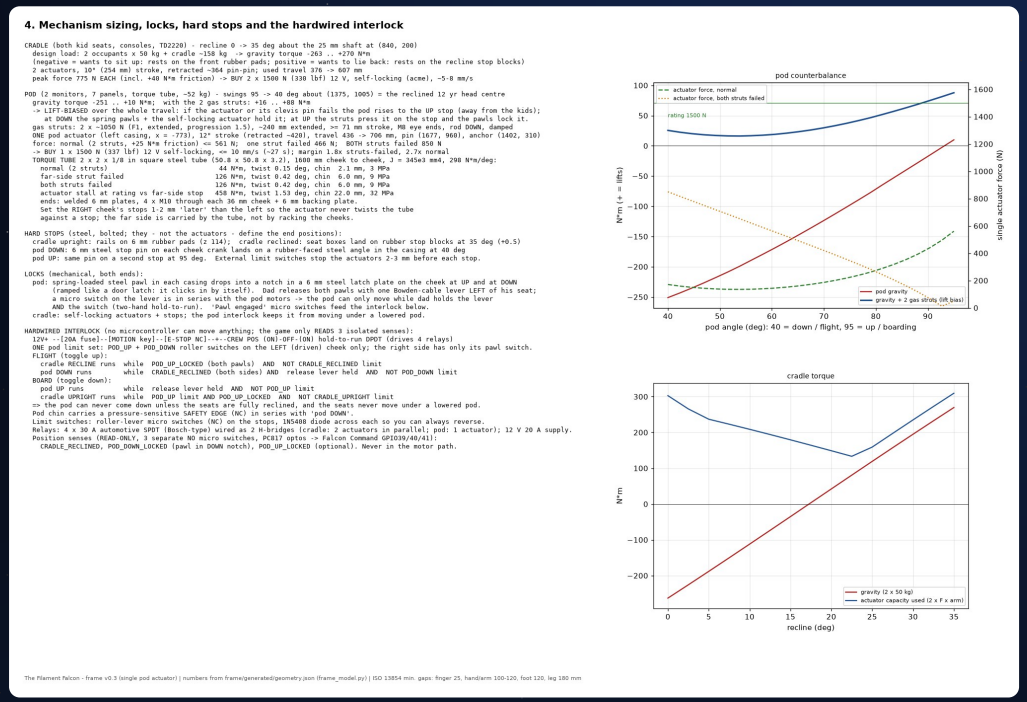
CRADLE · BOTH KID SEATS

Design load	258 kg (568 lb): 2 × 50 kg kids + frame + consoles
Gravity torque	-263 N·m (-194 lbf·ft) … +270 N·m (+199 lbf·ft): stable on both stops
Actuators	2 × 12 V, 10" stroke, pin-to-pin 376 → 607 mm, PUSH to recline
Peak force (model)	775 N (174 lbf) each → buy ≥ 1 500 N (330 lbf), self-locking

POD · MONITORS AND PANELS

Mass / CG radius	52 kg (115 lb) / 613 mm (2' 0.1") from the pivot
Swing	95° (parked up) → 40° (flight)
Gravity torque	-251 N·m (-185 lbf·ft) … +10 N·m (+7 lbf·ft)
2 gas struts	F1 1 050 N (236 lbf), progressive to 1575 N, 61 mm stroke used, ~240 mm extended
Net torque with struts	+16 N·m (+12 lbf·ft) … +88 N·m (+65 lbf·ft): rests firmly on whichever stop it is at
Actuators	2 × 12 V, 12" stroke, 436 → 706 mm, PULL to raise
Force (model)	566 N (127 lbf) each, 850 N (191 lbf) each if both struts fail → buy ≥ 1 000 N (225 lbf), self-locking

All values from `frame/generated/geometry.json`. Speed ≤ 10 mm/s (0.4 in/s) at the actuators, ~35-40 s per motion.



Drawing 04 · mechanism sizing: pod counterbalance (gravity vs. gravity + struts) and cradle torque vs. actuator capacity, plus hard stops, locks and the hardwired interlock.

POD MASS BUDGET (BESIDES THE WOOD)

- 2 monitors (C275B-144R 3.80 + C275W-1920RN 3.40 kg): 7.2 kg
- VESA adapters, rails, cables: 2.5 kg
- 7 printed panels + switches + 2 ESP32 boxes: 4.5 kg
- cheek steel: bearings, crank plates, latch plates: 6.0 kg
- foam + safety edge + trim: 1.5 kg

MOTION DRIVE PARTS from frame/BOM.md

QTY	ITEM	NOTES
2	**Cradle actuators**: 12 V DC linear actuator, 10" (254 mm) stroke, ≥ 1500 N (330 lbf), self-locking (acme/worm), ~364 mm (14.3") retracted pin-to-pin, ≤ 10 mm/s, internal limits	PUSH to recline; worst case ~775 N each (model). Retracted length must be 364 ±10 mm, else move the anchor in `frame_model.py` and re-run
1	**Pod actuator** (ONE, left casing): 12 V DC, 12" (300 mm) stroke, ≥ 1500 N (330 lbf), self-locking, 420 mm (16.5") retracted pin-to-pin, ≤ 10 mm/s, internal limits, 8 mm clevis holes. Chosen: RVMARINEPAT LA03-12-300-1500N (amazon.ca B0FKMVCQPC, CA$67.82 on 2026-10-10; same family as the cradle actuators)	extends to lower, retracts to raise; uses 436 → 706 mm of its 420 → 720; ≤ 561 N normally, 850 N if both struts fail (1.8x margin). A different part → edit POD_ACT_SPEC in `frame_model.py` and re-run
2	**Gas struts** (one per side), 10 mm rod / 22-28 mm body, ~240 mm extended, ≥ 71 mm stroke, F1 ~1050 N (235 lbf), progressive ~1.5, M8 eye (or M8 thread + eye) ends on 5 mm steel angle brackets with M8 grade 8.8 bolts (10 mm ball studs are only ~890 N rated), damped end of stroke	sized so the pod is LIFT-biased (+16..+88 N m): fail-safe UP if the actuator lets go. Buy 1000-1100 N, or an adjustable-valve 1200 N strut bled down. Mount rod DOWN, fit with the pod UP
1	**Torque tube**: 2" x 2" x 1/8" (50.8 × 50.8 × 3.2 mm) square steel tube (A500 / HSS), 1600 mm, with 2 welded 6 mm end plates 150 × 61 mm (4 × 11 mm holes each)	ties the right cheek to the driven left cheek (local metal shop; or no-weld: 2 × 50x50x5 steel angles per end)
2	6 mm steel backing plates 150 × 61 mm (4 × 11 mm holes)	outside face of each cheek at the tube end
8	M10 × 70 bolts, grade 8.8 + nylocks + washers	torque tube end plates through the 36 mm cheeks
3	Steel angle brackets 50 × 50 × 5 mm with M8 hole	2 strut anchors + 1 pod actuator anchor
2	6 mm steel latch plates (laser cut / filed from the cheek template `06`), with UP and DOWN notches	bolted to the cheeks
2	Spring-loaded steel pawls (gate latch / barrel bolt with spring)	engage the latch plates, one per casing
2	Bowden cable + lever (bicycle brake lever + inner/outer cable)	release both pawls from the lever box left of dad's seat
4	Steel stop pins 12 mm + rubber bumpers	pod UP / DOWN hard stops (on the cheeks, hitting steel angles)
2	Rubber stop blocks 10 mm on 2x4	cradle recline hard stop; 2 × 6 mm rubber pads for the upright stop

BEFORE ORDERING ACTUATORS

The geometry only works if the retracted pin-to-pin lengths match: cradle ≈ 364 mm (14.3"), pod ≈ 415 mm (16.3"), ±10 mm. Otherwise move the anchor in `frame/frame_model.py`, re-run it and re-check the clearances. Insist on **self-locking** (acme / worm) actuators: they are what holds the pod if a strut fails.

READ THIS CHAPTER BEFORE BUILDING ANYTHING THAT MOVES

04 SAFETY

A 44 kg (96 lb) pod swinging toward two children is the one part of this project that can really hurt someone. The answer is layered: **steel decides where things stop, struts decide what is heavy, and a person decides when anything moves**, with two hands, from the one seat that can see everyone. No microcontroller and no game code can move a seat or the pod.

1 · HARD STOPS DEFINE THE ENVELOPE

Steel stop pins/blocks at both ends of both motions; the actuators' own limits are 2-3 mm beyond the external limit switches, and the stops are where the clearances above were measured.

2 · NEAR-NEUTRAL POD

The struts more than cancel gravity everywhere (net +16 .. +88 N m), so: actuator / clevis pin / bracket fails mid-travel → the pod rises to the UP stop (strut damping, rubber stop) and the UP pawls catch it when dad lets go of the lever; at DOWN the spring pawls (both sides) hold it down; at UP the struts press it on the stop and the pawls lock it. One strut fails → the self-locking actuator (1500 N vs 466 N needed) holds; both struts fail → still 1.8x margin. Only a double failure (both struts AND the actuator) lets the pod come down - onto the steel DOWN stops, which keep it 195 mm off the legs.

3 · SLOW

≤ 10 mm/s actuators, ~35-40 s per motion.

4 · MECHANICAL LOCKS AT BOTH ENDS + TWO-HAND CONTROL

Spring pawls in each casing drop into notches in a 6 mm steel latch plate on each cheek (UP and DOWN). They click in by themselves; a Bowden-cable lever LEFT of dad's seat lifts both. A micro switch on the lever is in series with the pod motors, so the pod moves only while dad holds the lever with one hand AND the hold-to-run toggle with the other (two-hand control). The pod has ONE set of limit switches (POD_UP, POD_DOWN on the driven left cheek); the right cheek only has its pawl switch.

INTERLOCK, KEY, E-STOP AND TWO-HAND CONTROL

HARDWIRED INTERLOCK: WHAT MAY MOVE, AND WHEN

MOTION	DAD HOLDS	ONLY WHILE (LIMIT / LOCK SWITCHES)
Cradle RECLINE	FLT	POD_UP_LOCKED (both pawls), not yet at the CRADLE_RECLINED limit
Pod DOWN	FLT **+ release lever**	CRADLE_RECLINED (both sides), not at POD_DOWN; safety edge not pressed
Pod UP	BRD **+ release lever**	not at the POD_UP limit
Cradle UPRIGHT	BRD	POD_UP limit AND POD_UP_LOCKED, not at CRADLE_UPRIGHT
Anything	-	MOTION key on, E-stop released, 20 A fuse intact

- FLIGHT: cradle reclines only while POD_UP_LOCKED; pod comes down only while CRADLE_RECLINED (both sides) AND lever held.
- BOARD: pod goes up while lever held; cradle comes upright only while POD_UP AND latched.
- ⇒ the pod can never come down unless the seats are fully reclined, and the seats never move under a lowered pod.
- Key switch, 20 A fuse, latching E-stop (NC), NC roller limit switches each bridged by a 1N5408 so you can always drive back.
- Pressure-sensitive safety edge (NC) along the pod chin in series with "pod DOWN".
- READ-ONLY position senses for the game: **3** separate micro switches (CRADLE_RECLINED, POD_DOWN_LOCKED = pawl in the DOWN notch, POD_UP_LOCKED - optional), each through a PC817 optocoupler (2.2 kOhm 1/2 W on the 12 V side, 10 kOhm pull-up to 3V3) to Falcon Command GPIO39 / 40 / 41. They are not in the motor path; the game only shows the state and calls "pod coming down" on the intercom.

CONTROLS AT DAD'S SEAT ONLY (CMD_B + LEVER BOX)

- **MOTION key** (key removable in OFF only) drives a 30 A relay: key out = no motion.
- **22 mm mushroom E-stop**, NC, latching, twist-release.
- **CREW POS** heavy DPDT (ON)-OFF-(ON) toggle, ≥ 15 A at 12 V: FLT / BRD, hold-to-run. Let go = stop.
- **Pod release lever** (Bowden cable, left of the seat) lifts both pawls and closes a micro switch in series with the pod motors.
- **20 A blade fuse** at the 12 V 20 A supply; 4 × 30 A automotive relays as two H-bridges.

SOFTWARE ONLY READS

Falcon Command reads separate dry contacts (cradle reclined, pod down) so the game can show the state. It never drives the actuators.

CLEARANCES AND PINCH POINTS

SMALLEST POD-TO-CHILD GAPS OVER THE WHOLE SWING

KID	BODY PART	POD PART	POD ANGLE	MIN. GAP
PILOT	upper arm	Pod cheek L	95°	**155 mm** (6.1")
PILOT	thigh	Pod chin plate	40°	**195 mm** (7.7")
PILOT	shin	Pod chin plate	40°	**220 mm** (8.7")
PILOT	torso	Pod cheek L	95°	**233 mm** (9.2")
OPS	upper arm	Pod cheek R	95°	**241 mm** (9.5")
PILOT	head	Pod cheek L	40°	**251 mm** (9.9")
OPS	head	Pod cheek R	77°	**261 mm** (10.3")

SMALLEST CRADLE-TO-STRUCTURE GAPS OVER THE RECLINE

MOVING	FIXED	RECLINE	MIN. GAP
Cradle rail L	Nose box rear wall	7.5°	**34 mm**
PILOT kid upper arm	Casing front wall L (inner)	32.5°	**35 mm**
PILOT kid upper arm	Casing inner wall L	35.0°	**35 mm**
Cradle rail R	Nose box rear wall	5.0°	**35 mm**
Cradle toe lip	Nose box rear wall	0.0°	**40 mm**

Worst case = 12-year-old manikin, checked by `frame/frame_model.py` at every pod and cradle angle.

PINCH-POINT RULES (ISO 13854 STYLE)

every gap that opens/closes is either ≥ 120 mm (head/body) or ≤ 8 mm / covered (finger can't enter) or ≥ 25 mm at all times. The cheeks run in 48 mm slots in the casings closed with brush strips; a fixed 120 mm crush-zone ramp sits under the reclined seat backs so nothing can be trapped between seat back and floor; the nose-box actuator slots have brush strips. See `drawings/05_pinch_points_clearances`.



Drawing 05 · every gap that changes with motion, checked against finger 25 / head 120 mm.

HARNESS AND LOAD PATH



Harness hub, lid off. The kit's own female buckle bodies sit in fixed pockets: shoulders 30° from vertical, lap straps from the sides, crotch/anchor strap out the bottom.

PRINTING THE HUB

Print: **PETG** (not PLA: brittle when cold, creeps), 0.2 mm layers, 6 perimeters, 5 top / 5 bottom layers, 40-50 % gyroid, base floor-down, lid flat, no supports.

LOAD PATH: THE PRINT CARRIES NO STRAP LOAD

strap → tongue → the kit's female body → its webbing bar → a short sewn webbing loop → **welded steel O-ring** (50 mm OD, 6 mm wire) inside the hub → crotch/anchor strap looped around the same ring → forged eye bolt in the cradle. If the hub broke, the straps would still be joined through the webbing and the ring.

PLASTIC-BUCKLE WARNING: POSITIONING HARNESS ONLY

This is a positioning harness for a slow-moving home cockpit (kids sit in a reclining chair, nothing accelerates), **not** a crash- or fall-rated restraint. The kit's plastic (POM) side-release buckles are luggage / backpack hardware; sellers quote only ~50 kg breaking load and they are not certified for life-safety use. Never use this harness for anything that can fall, tip or accelerate (no swings, no vehicles, no hanging). Sew loops with bar-tacks in polyester thread (or use rated metal buckles), inspect the buckles for cracks monthly and replace on any damage. Kids must be able to release themselves: test that each child can squeeze the side-release on all four buckles.

A rated metal-buckle 4-point harness is listed as an ALTERNATIVE row in the BOM ($111.98, not in the totals). Anchor every harness with M10 (3/8") forged eye bolts through the cradle rails or seat-back frames, never plywood alone.

RULES, CHECKS AND THE HOLD-TO-RUN CIRCUIT

SAFETY RULES · PUT A COPY ON THE CMD CONSOLE

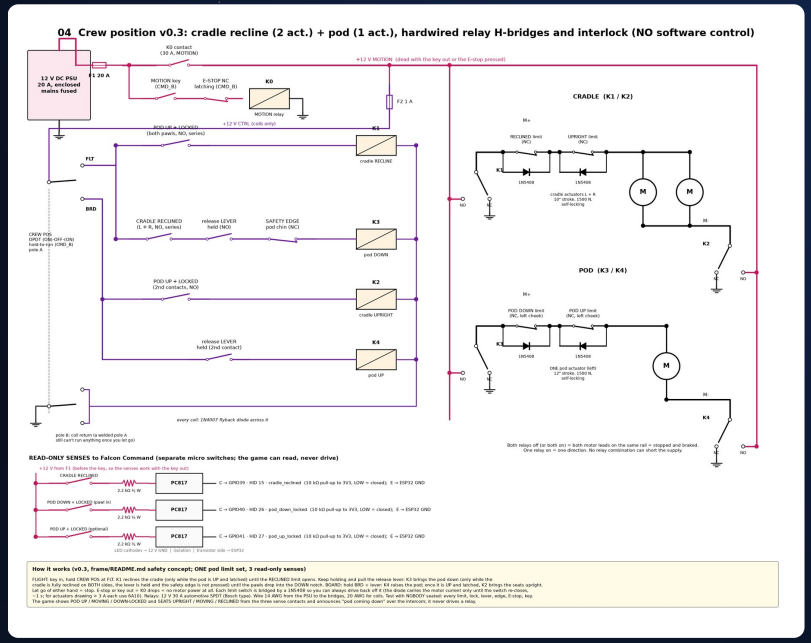
- Motion ONLY from dad's seat, hold-to-run + release lever (two hands). Let go = stop. Key out when not flying.
- Harnesses fastened, hands on lap, heads on the headrests BEFORE "FLT". The game checklist shows it; honour system.
- Nobody stands beside the casings or in front of the nose while anything moves.
- Monthly: strut force (with the actuator clevis pin pulled and the pawls released, the pod must rise by itself to the UP stop; if it stays or sinks, replace the struts before the next flight), torque-tube bolts, actuator clevis pins + R-clips, pawls/notches, stop pins, safety cables on the monitors, harness webbing and eye bolts.

7 · CABLES ACROSS PIVOTS

Every bundle crossing a pivot gets two printed `pivot_cable_clamp`s (fixed side + moving side, ≤ 60 mm from the axis) with a 200 mm U service loop in braided sleeve.

FIRST POWER-UP OF THE MOTION SYSTEM

Test with **nobody seated**: every limit, every lock, the E-stop, the key, lever released = no pod motion, safety edge = stop. Then re-check every gap on drawing 05 with a 25 mm (1") dowel and a 120 mm (4.7") ball at both ends of both motions.

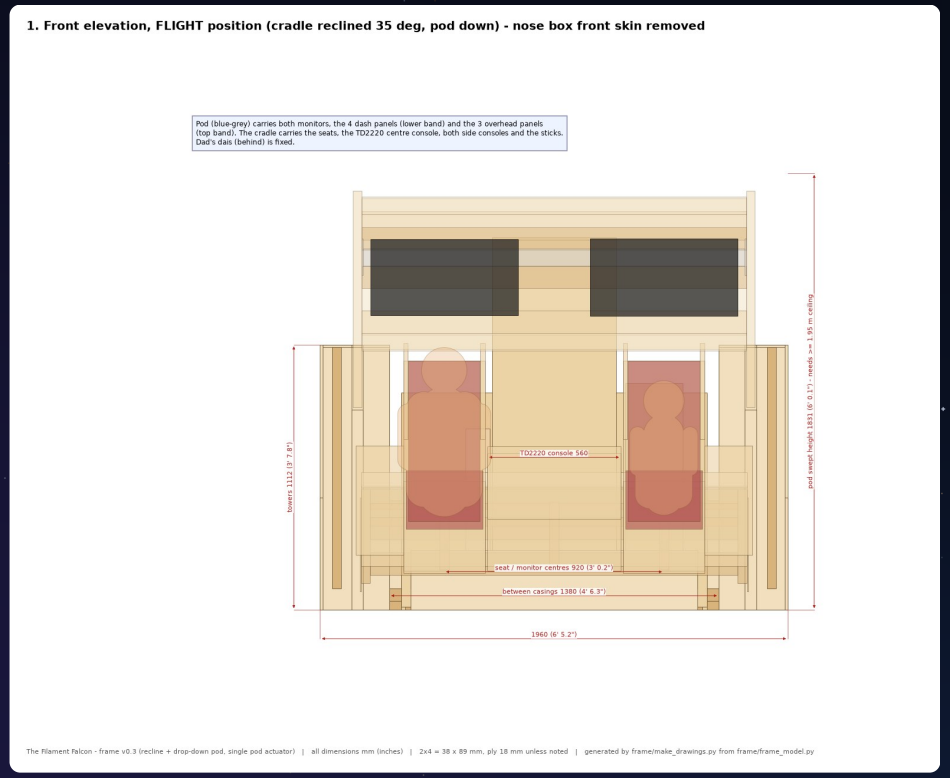


Electronics diagram 04 · crew-position circuit v0.2. 20 A fuse, key + E-stop drop the MOTION relay, CREW POS hold-to-run feeds the relay coils only through the interlock (pod up + locked, cradle reclined L + R, release lever, pod-chin safety edge), 4 relays as two H-bridges (cradle, pod), NC limit switches with 1N5408 drive-back diodes, and three isolated read-only senses (cradle reclined, pod down + locked, optional pod up + locked) for Falcon Command.

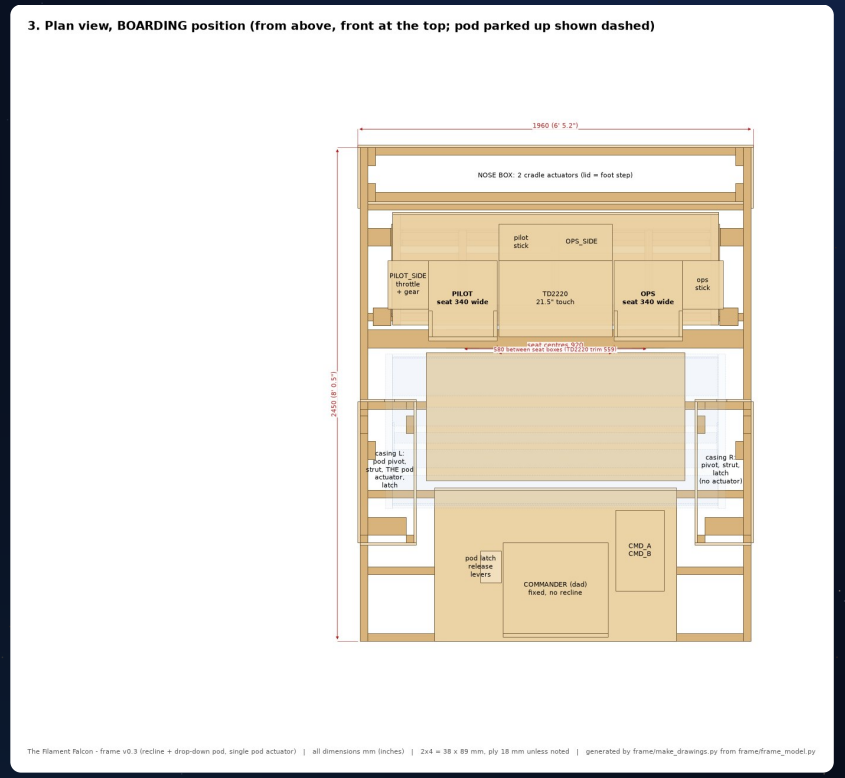
2X4 LUMBER AND PLYWOOD ON A FLOOR-LEVEL BASE

05 WOODEN FRAME

2x4 lumber + 12/18 mm plywood on a floor-level base (no raised platform). Two kid seats (6-12 yr) side by side in front, dad's seat behind and centred on a 250 mm dais. All dimensions mm (inches in brackets). 2x4 = 38 × 89 mm (1.5" x 3.5").

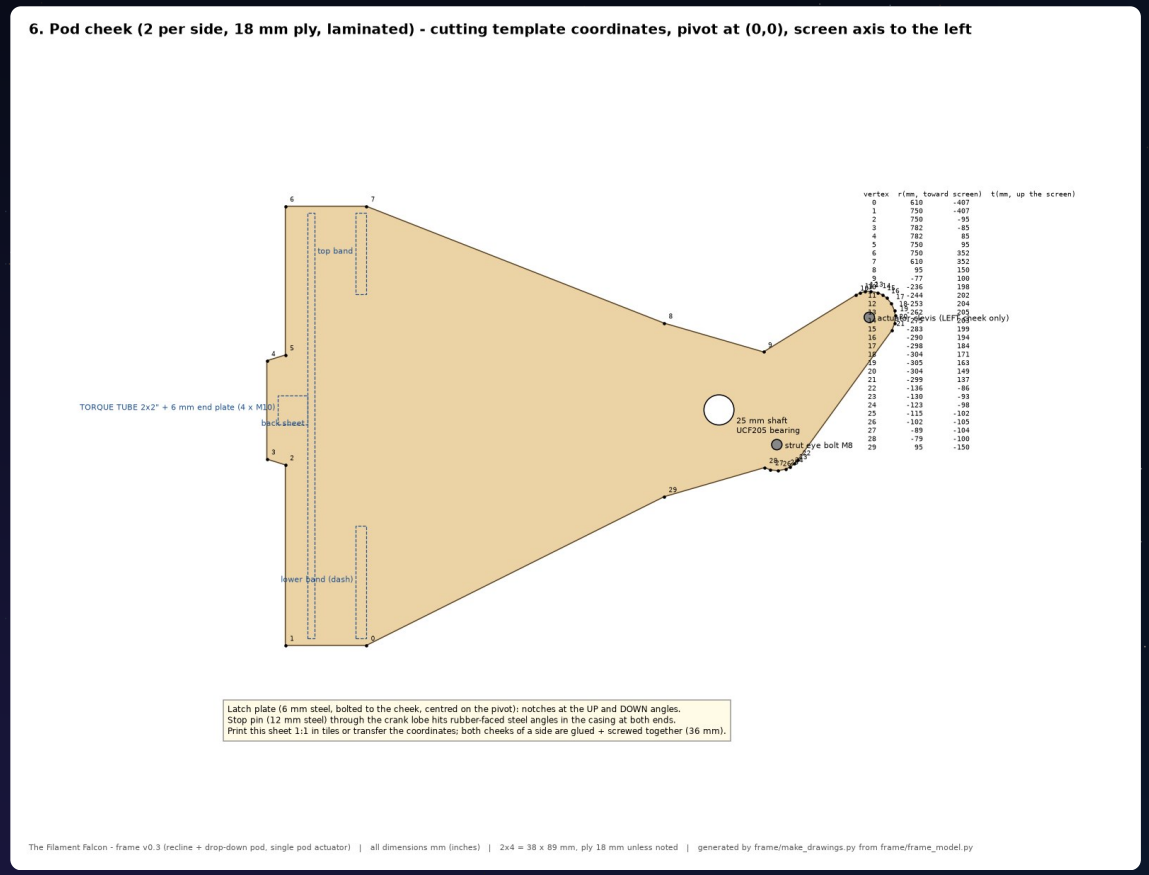


Drawing 01 · front elevation in FLIGHT: casings, pod, seats and monitor positions.

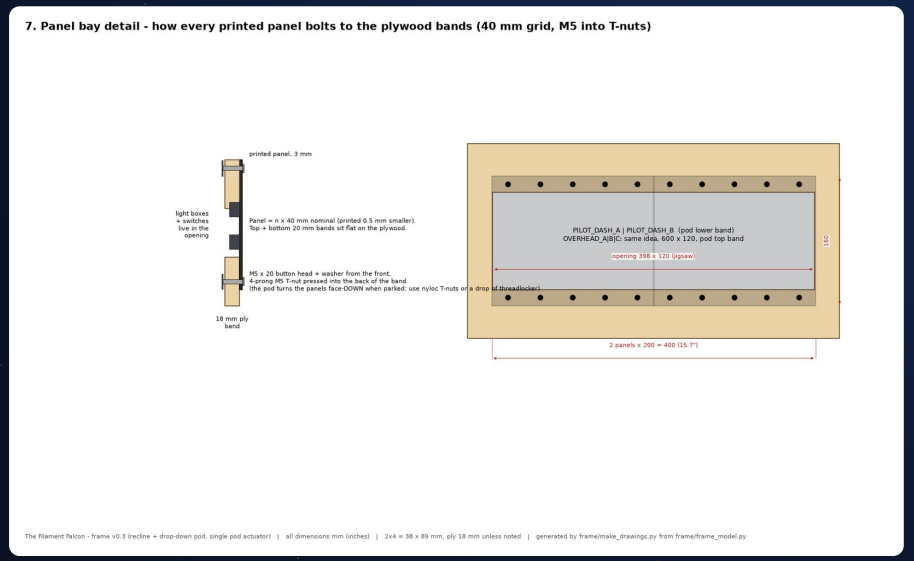


Drawing 03 · plan: seats at x = ±460 mm, consoles, casings, actuators, Dad's dais.

TEMPLATES AND PANEL BAYS



Drawing 06 · pod cheek template, full size (print tiled, trace onto plywood). Two layers of 18 mm ply per cheek, glued, with steel crank and latch plates bolted through.



Drawing 07 · panel bay detail: how a printed panel bolts to the plywood rails on the 40 mm grid with M5 T-nuts.

CUT LIST SUMMARY

CUT LIST BY GROUP AND MATERIAL

GROUP	MATERIAL	PIECES	TOTAL LENGTH / AREA
fixed	2x4 (38 x 89)	34	39.9 m (131 ft) of 2x4
fixed	12 mm ply	25	7.53 m² (81 sq ft)
fixed	18 mm ply	6	4.65 m² (50 sq ft)
fixed	18 mm ply box	2	0.33 m² (4 sq ft)
cradle	2x4 (38 x 89)	9	8.4 m (28 ft) of 2x4
cradle	18 mm ply	7	1.84 m² (20 sq ft)
cradle	12 mm ply	1	0.11 m² (1 sq ft)
cradle	6 mm ply/hardboard	2	0.15 m² (2 sq ft)
cradle	18 mm ply box	6	0.67 m² (7 sq ft)
pod	12 mm ply	3	1.51 m² (16 sq ft)
pod	18 mm ply	4	2.69 m² (29 sq ft)
pod	2x4 (38 x 89)	2	3.2 m (11 ft) of 2x4
All		**101**	

2x4 boards: 45 pieces from **22 × 8 ft (2438 mm)** + **2 × 10 ft (3048 mm)** boards (first-fit packing, 4 mm kerf; buy 3 spares). The 2450 mm sills need 10 ft boards (or 2 × 8 ft spliced over a base cross).

- 12 mm ply: 9.16 m2 (99 sq ft) → **4 sheet(s)**
- 18 mm ply: 13.48 m2 (145 sq ft) → **6 sheet(s)**
- 6 mm ply/hardboard: 0.15 m2 (2 sq ft) → **1 sheet(s)**

Summarised from `frame/cut_list.csv`; every piece with its notes is in `frame/cut_list.md`. Groups: fixed = base, nose box, towers/casings, ramp, Dad's dais; cradle = reclining kid-seat frame; pod = rotating instrument pod.

LUMBER AND SHEET GOODS TO BUY

QTY	ITEM	NOTES
24	2x4 stud, 2438 mm (8 ft), straight, kiln dried	22 used by the packing in `cut_list.md` + 2 spares (38 × 89 mm actual)
2	2x4, 3048 mm (10 ft)	the two 2450 mm sills
6	18 mm (3/4") plywood, 1220 × 2440 mm	seat boxes/backs, consoles, cradle deck, pod cheeks (2 layers each), pod bands, dais (13.5 m2 net)
4	12 mm (1/2") plywood, 1220 × 2440 mm	casings, skins, nose box, pod back sheet + chin, crush-zone ramp (9.2 m2 net)
1	6 mm hardboard, 1220 × 2440 mm	small covers / templates

MEASURE YOUR LUMBER FIRST

2x4 = 38 × 89 mm (1.5" × 3.5") actual. Check real sizes and adjust `frame_model.py` before cutting; re-running it regenerates the cut list, the drawings and this book.

ASSEMBLY STEPS from frame/README.md

1. **Cut** everything in `cut_list.md`; label each piece. Check real lumber sizes and adjust `frame_model.py`.
2. **Base**: sills + cross members (the nose-rear and cradle-front crosses are laid FLAT so the actuators pass over them). Square the diagonals. Build it where it will live.
3. **Nose box**: posts, front skin, rear wall with 2 actuator slots + brush strip, screwed-down lid (foot step).
4. **Towers and casings**: outer + inner tower posts each side carry the pod shaft (25 mm, UCF205 4-bolt flange bearings on 6 mm steel plates). Casing walls get the 48 mm cheek slot cut from the template (`06`), edged with brush strip.
5. **Cradle**: two rails on the 25 mm cradle shaft (2 x UCP205 pillow blocks on the base), seat boxes and backs (harness slots), side consoles, centre console with the TD2220 shelf. Fit the stop rail + rubber blocks.
6. **Cradle actuators** in the nose box (x = ±560, anchor 20 mm behind the nose, 30 mm up; pin on the cradle 360 mm back, 190 mm up). Bench-set both to the same length before fitting. Test the recline empty first.
7. **Pod**: two cheeks (2 layers 18 mm ply, glued, steel crank/latch plates bolted through), back sheet, dash lower band, overhead upper band. Hang it on the shaft with the pod parked UP and the pawls engaged.
1. **Torque tube**: bolt the 2x2x1/8" tube (welded 6 mm end plates) between the cheeks behind the back sheet, 4 x M10 through each 36 mm cheek with a 6 mm backing plate outside. Check the cheeks are parallel and the pod turns freely on both bearings before anything else is fitted.
2. **Gas struts** (rod down, damped): fit with the pod UP and locked. With the actuator NOT fitted, the pod must rise by itself from any position to the UP stop (slowly, damped) - that is the fail-safe direction. Pushing it down by hand takes up to ~11 kg (107 N) at the chin - expected.
3. **The pod actuator** in the LEFT casing only; set its length with the pod on the UP stop. Set the RIGHT cheek's stop pins 1-2 mm 'later' than the left ones so the actuator never twists the torque tube against a stop.
4. **Interlock wiring** per `drawings/04_mechanism_interlock`. Test with NOBODY seated: every limit, every lock, E-stop, key, lever released = no pod motion, safety edge = stop.
5. **Guards and skins**: brush strips, casing skins, crush-zone ramp, nose lid. Re-check every gap on `05` with a 25 mm dowel and a 120 mm ball at both ends of both motions.
6. **Monitors, panels, harness**: see `docs/mechanical/README.md` and `docs/mechanical/displays.md`.

BACKLIT LETTERING, LIGHT BOXES, ONE MOUNTING GRID

06 3D-PRINTED PANELS

BACKLIT THROUGH-LETTERS

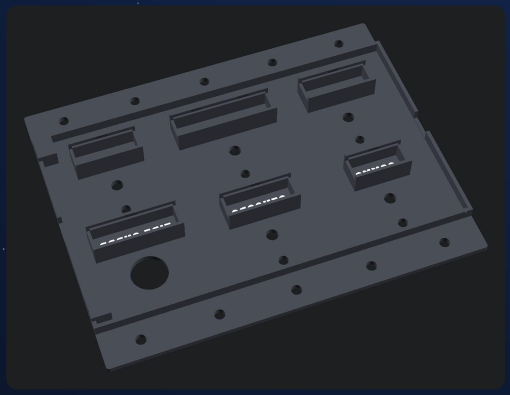
All panels are 3 mm thick, dark body, white letters that go through the full thickness. Panels are multiples of the 40 mm grid and printed 0.5 mm smaller than nominal so neighbours never bind. The top and bottom 20 mm bands carry the M5 holes (10 mm from the edge, every 40 mm) and sit flat on the plywood rails; all light boxes, ribs and switch bodies are inside the 120 mm (or 80 / 160 mm) bay opening. Lettering is Barlow Condensed Black (SIL OFL), from `cad/fonts/`.

LED LIGHT BOXES ON THE BACK

Behind every label is a light box (1.6 mm walls, opaque) about 12.8 mm deep: the 12 mm WS2812B strip (10 mm fits too) clicks in behind small ledges at 9 mm from the panel face (stand-off for even light), with snap nubs to hold it. Each box end wall has a notch for the strip or a 3-wire jumper. ONE strip snakes through all panels in the order of `panel_order` in `shared/panel_labels.json` (CMD_A → CMD_B → PILOT_SIDE → OPS_SIDE → OPS_DASH_A → OPS_DASH_B → OVERHEAD_C → OVERHEAD_B → OVERHEAD_A → PILOT_DASH_A → PILOT_DASH_B); inside a panel, labels are visited row by row in a snake. Between boxes and between panels, solder short 3-wire jumpers (strip cut on the copper pads). The perimeter rib has wire notches. Real LED counts per label are measured later into `shared/led_segments.json`.



Front: dark body, white letters through the full 3 mm.



Back: one light box per label, strip ledges and wire notches.

MOUNTING GRID: HOW A PANEL BOLTS TO THE FRAME

1. The bay in the 18 mm plywood is an opening (e.g. 400 × 120 mm for the dash) with rails above and below. See `frame/drawings/07_panel_bay_detail`.
2. Drill holes for the T-nut barrel (usually 7 mm; check yours) on the 40 mm grid, 10 mm from the opening edge, and press M5 four-prong T-nuts in from the BACK.
3. Fit the switches into the panel first (the switch nuts clamp on 3 mm), plug the LED strip into its boxes, then put the panel in the bay and bolt with M5 × 20 button-head screws + washers (M5 × 16 if the T-nut barrel is long).
4. Split panels (A + B) join with two 3 mm x 16 mm pins (`alignment_pins_x8.3mf`, or 3 mm steel dowels) in the seam bosses.
5. Prefer 1/4-20? Set `FF_M5_D = 6.9` in `ff_lib.scad` and rebuild.

LIGHT-LEAK CHECK

Light leak check: print black PETG/PLA at ≥ 3 walls; if light glows through the body, paint the inside of the light boxes black or use a "blackout" filament.

ALL ELEVEN PANELS one multi-material file each

PANEL	SIZE	LABELS	WHERE
PILOT_DASH_A	200 × 160	7	Pilot dash (pod lower band), outboard half (left seat)
PILOT_DASH_B	200 × 160	8	Pilot dash, inboard half (joins A with 2 alignment pins)
PILOT_SIDE	200 × 160	6	Pilot outboard side console: THROTTLE + LANDING GEAR slots
OPS_DASH_A	200 × 160	6	Ops dash, inboard half: power toggles
OPS_DASH_B	200 × 160	8	Ops dash, outboard half: mode rotary + payload
OPS_SIDE	200 × 160	4	Ops inboard side console: JUMP ARM (guarded) + ENGAGE (collar)
OVERHEAD_A/B/C	3 × 200 × 120	3/3/4	pod upper band (above the monitors), launch checklist row + lights + EJECT
CMD_A	200 × 200	8	commander: master power key, mission, start/pause, caution reset, PA, dimmer
CMD_B	200 × 200	10	commander: failure injection + crew **MOTION** (key, FLT/BRD hold-to-run, E-stop)

70 backlit labels in total, each listed with id, panel, text, position and strip order in `shared/panel_labels.json`. The LED strip visits the panels in this order:

CMD_A → **CMD_B** → **PILOT_SIDE** → **OPS_SIDE** → **OPS_DASH_A** → **OPS_DASH_B** → **OVERHEAD_C** → **OVERHEAD_B** → **OVERHEAD_A** → **PILOT_DASH_A** → **PILOT_DASH_B**

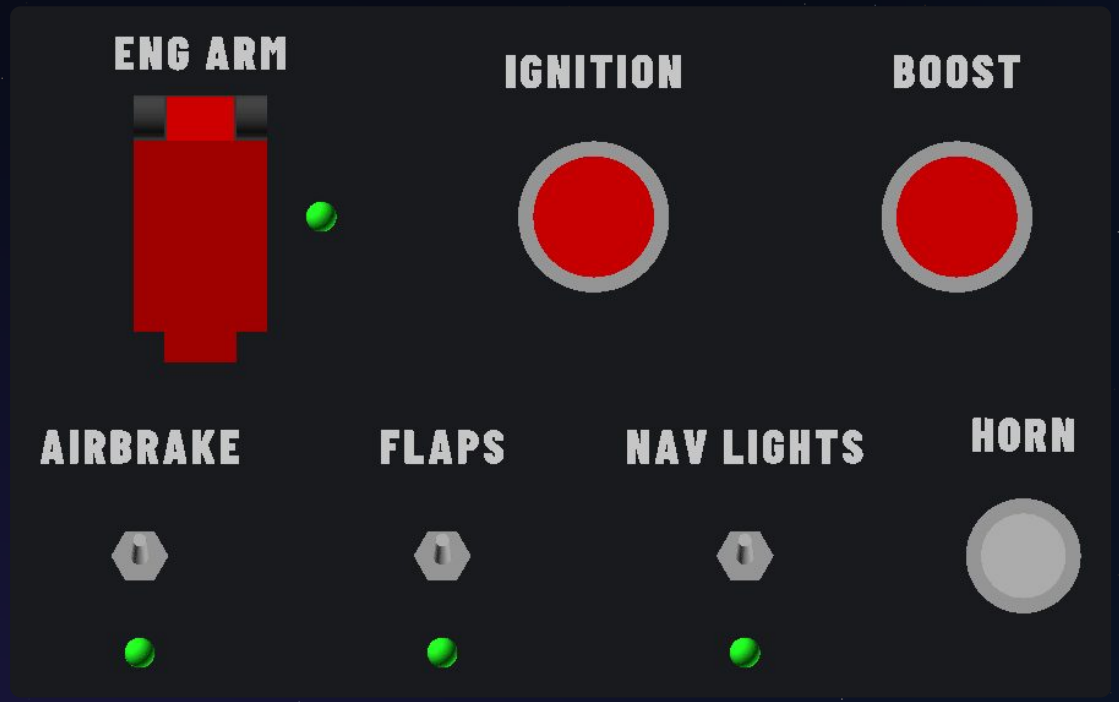
ONE MULTI-MATERIAL FILE

`<ID>.3mf` - one multi-material file, two objects. **panel** is extruder 1, black (`#1E1F22`). **letters** is extruder 2, white (`#F2F2F2`): the lettering through the full 3 mm face, plus a 0.8 mm box behind each label. That box is the old letter-insert plaque (rounded, inset 0.3 mm so it sits inside the light box) and it is the diffuser the LEDs shine on. The two objects stay aligned; they do not overlap. Opens with both filaments and both extruders assigned in PrusaSlicer, OrcaSlicer and Bambu Studio.

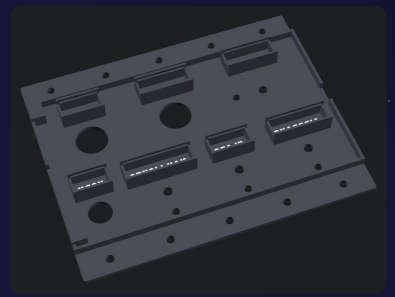
PNG: `cad/export/png/panels/<ID>_front.png` and `<ID>_back.png`.

PANEL PILOT_DASH_A

Pilot dash, left piece



PILOT_DASH_A · front. Pilot dash, left piece.



Back: light boxes behind every label.

Nominal size	200 × 160 mm (7.9" × 6.3")
Labels / LEDs	7 labels · 18 LEDs (placeholder map)
Print volume	116 cm³ (body + letters)
3MF check	**watertight, no problems**

WHERE IT GOES

Pod lower band (drops down in front of the reclined crew), in front of the LEFT seat, outboard half.

CONTROLS

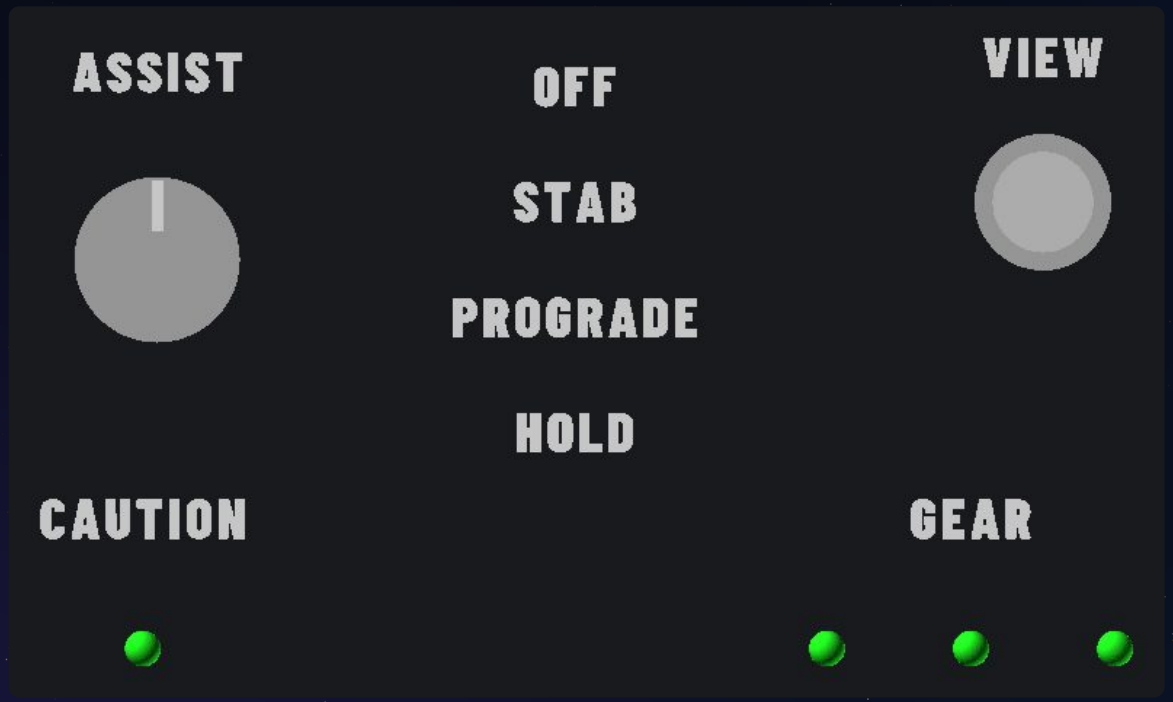
QTY	PART	USED FOR
2	22 mm push button, 22 mm panel hole (industrial 22 mm or mini arcade)	BOOST, IGNITION
1	16 mm momentary push button (metal/illuminated ok)	HORN
4	5 mm LED + bezel/holder (or a WS2812B 5 mm through-hole)	AIRBRAKE status, ENG ARM status, FLAPS status, NAV LIGHTS status
3	Mini toggle switch SPST/SPDT, 6 mm (1/4") bushing (MTS-102 style)	AIRBRAKE, FLAPS, NAV LIGHTS
1	Mini toggle 6 mm + printed missile guard (toggle_guard.3mf)	ENG ARM

BACKLIT LABELS (LED ESTIMATE AT 60/M)

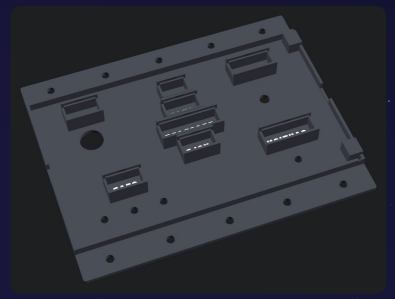
ENG ARM 2 · IGNITION 2 · BOOST 2 · HORN 1 · NAV LIGHTS 3 · FLAPS 2 · AIRBRAKE 2

PANEL PILOT_DASH_B

Pilot dash, right piece



PILOT_DASH_B · front. Pilot dash, right piece.



Back: light boxes behind every label.

Nominal size	200 × 160 mm (7.9" × 6.3")
Labels / LEDs	8 labels · 15 LEDs (placeholder map)
Print volume	118 cm³ (body + letters)
3MF check	**watertight, no problems**

WHERE IT GOES

Pod lower band in front of the LEFT seat, inboard half.

CONTROLS

QTY	PART	USED FOR
1	16 mm momentary push button (metal/ illuminated ok)	VIEW
1	EC11 rotary encoder with push switch + knob	
4	5 mm LED + bezel/holder (or a WS2812B 5 mm through-hole)	MASTER CAUTION (red), gear down (green)

BACKLIT LABELS (LED ESTIMATE AT 60/M)

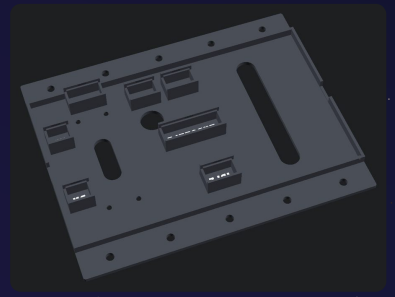
ASSIST 2 · VIEW 1 · OFF 1 · STAB 1 · PROGRADE 2 · HOLD 1 · GEAR 1 · CAUTION 2

PANEL PILOT_SIDE

Pilot side console (throttle + gear)



PILOT_SIDE · front. Pilot side console (throttle + gear).



Back: light boxes behind every label.

Nominal size	200 × 160 mm (7.9" × 6.3")
Labels / LEDs	7 labels · 8 LEDs (placeholder map)
Print volume	107 cm³ (body + letters)
3MF check	**watertight, no problems**

WHERE IT GOES

Outboard LEFT side console of the reclining cradle, horizontal, top of panel = forward.

CONTROLS

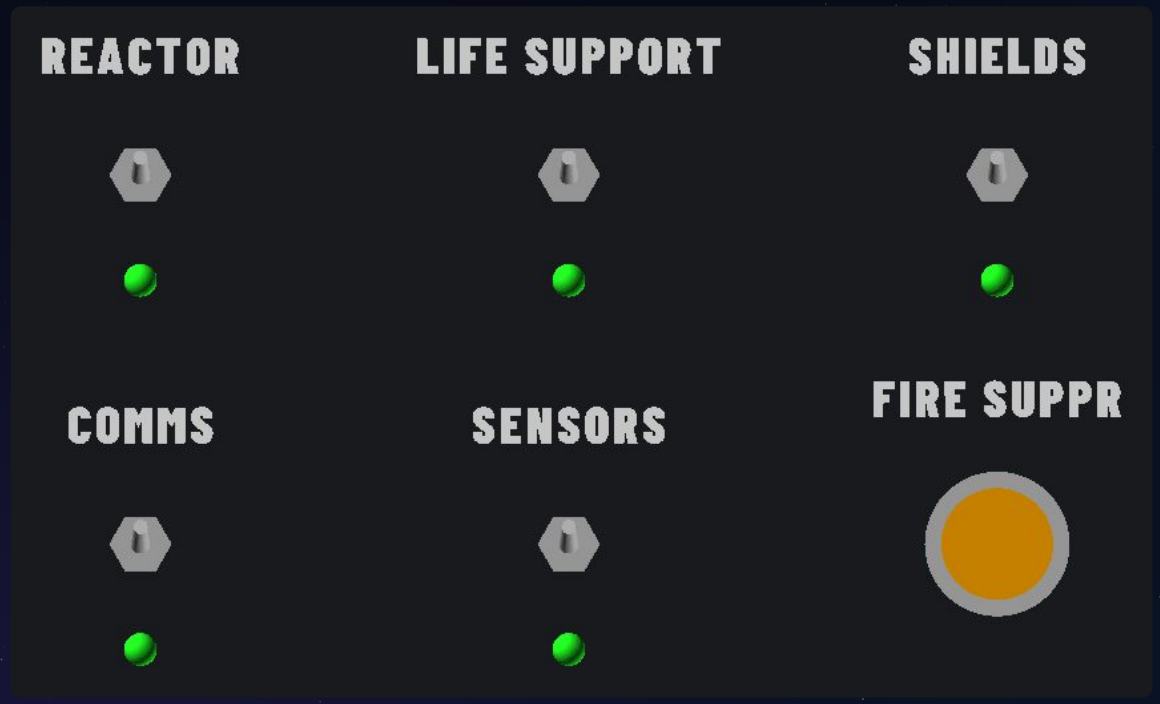
QTY	PART	USED FOR
1	16 mm momentary push button (metal/ illuminated ok)	PTT
2	lever slot (printed mechanism below the panel)	LANDING GEAR lever slot (gear lever box mounts below), THROTTLE lever slot (throttle quadrant mounts below)

BACKLIT LABELS (LED ESTIMATE AT 60/M)

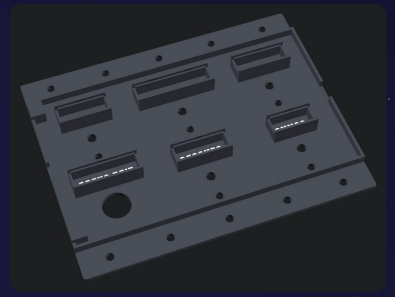
GEAR 1 · PTT 1 · MAX 1 · UP 1 · THROTTLE 2 · DN 1 · IDLE 1

PANEL OPS_DASH_A

Ops dash, left piece (power)



OPS_DASH_A · front. Ops dash, left piece (power).



Back: light boxes behind every label.

Nominal size	200 × 160 mm (7.9" × 6.3")
Labels / LEDs	6 labels · 19 LEDs (placeholder map)
Print volume	117 cm³ (body + letters)
3MF check	**watertight, no problems**

WHERE IT GOES

Pod lower band in front of the RIGHT seat, inboard half.

CONTROLS

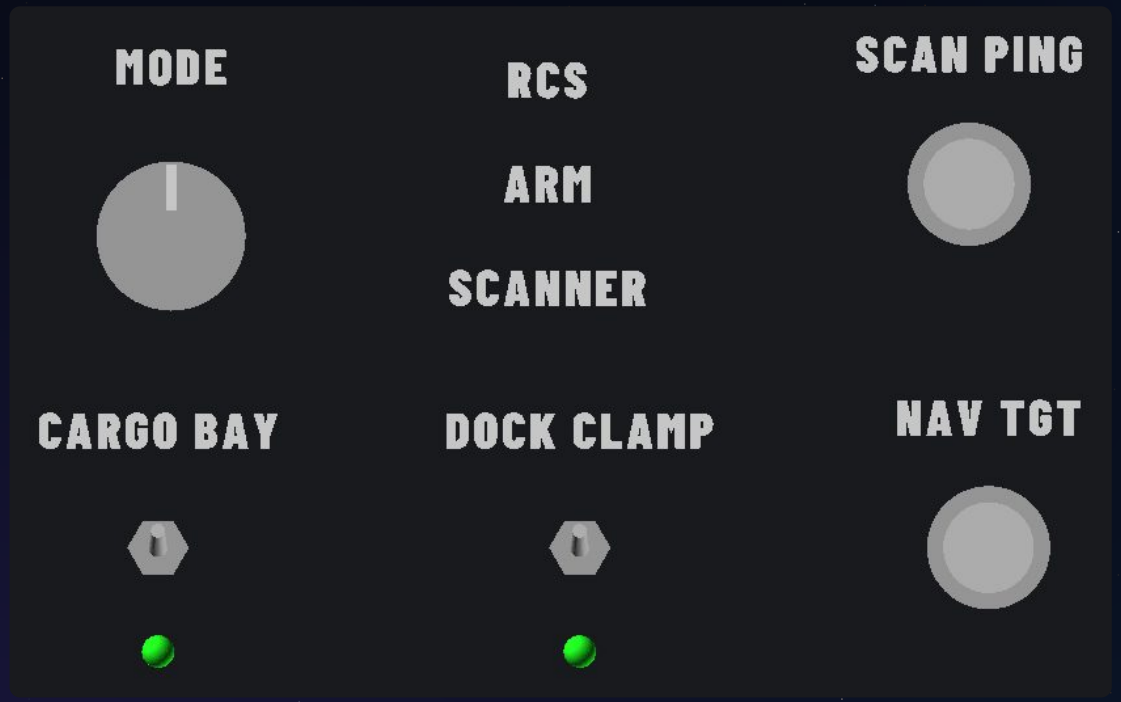
QTY	PART	USED FOR
1	19 mm push button, 19 mm panel hole (metal anti-vandal or mini arcade)	FIRE SUPPR
5	5 mm LED + bezel/holder (or a WS2812B 5 mm through-hole)	COMMS status, LIFE SUPPORT status, REACTOR status, SENSORS status, SHIELDS status
5	Mini toggle switch SPST/SPDT, 6 mm (1/4") bushing (MTS-102 style)	COMMS, LIFE SUPPORT, REACTOR, SENSORS, SHIELDS

BACKLIT LABELS (LED ESTIMATE AT 60/M)

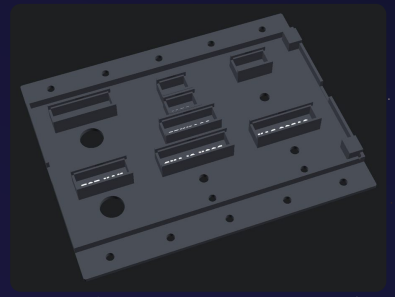
REACTOR 2 · LIFE SUPPORT 3 · SHIELDS 2 · FIRE SUPPR 3 · SENSORS 2 · COMMS 2

PANEL OPS_DASH_B

Ops dash, right piece (mode + payload)



OPS_DASH_B · front. Ops dash, right piece (mode + payload).



Back: light boxes behind every label.

Nominal size	200 × 160 mm (7.9" × 6.3")
Labels / LEDs	8 labels · 17 LEDs (placeholder map)
Print volume	120 cm³ (body + letters)
3MF check	**watertight, no problems**

WHERE IT GOES

Pod lower band in front of the RIGHT seat, outboard half.

CONTROLS

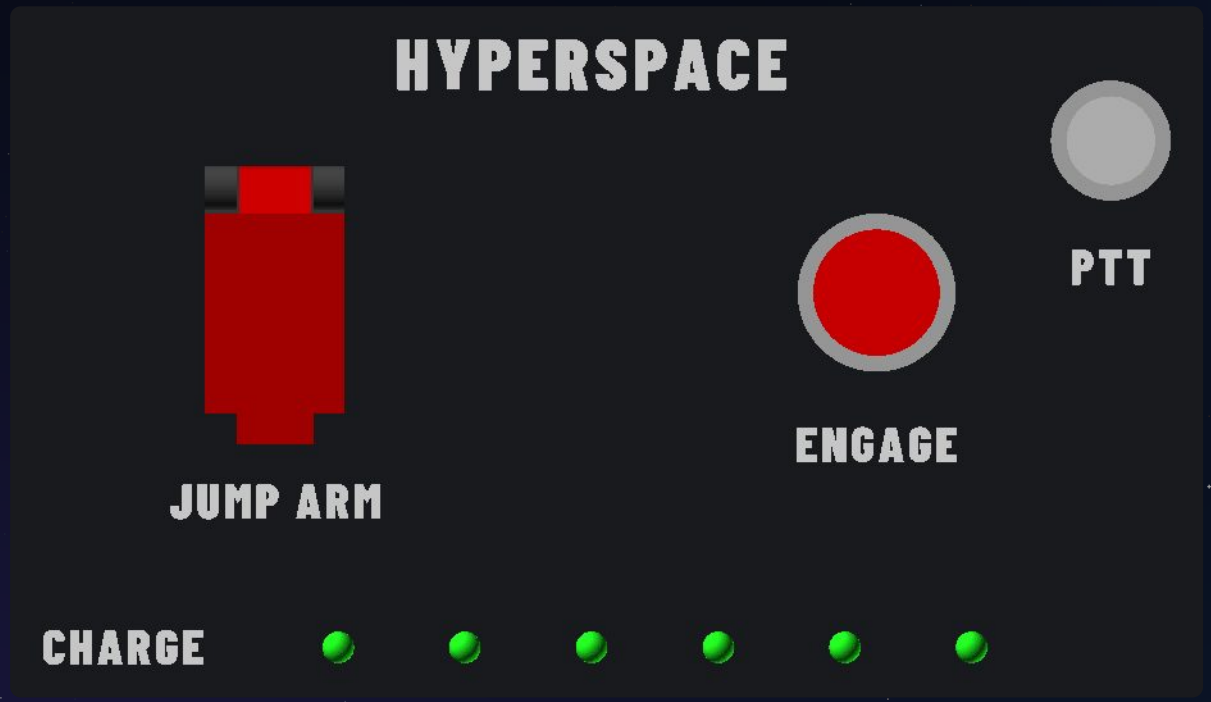
QTY	PART	USED FOR
2	16 mm momentary push button (metal/ illuminated ok)	NAV TGT, SCAN PING
1	EC11 rotary encoder with push switch + knob	
2	5 mm LED + bezel/holder (or a WS2812B 5 mm through-hole)	CARGO BAY status, DOCK CLAMP status
2	Mini toggle switch SPST/SPDT, 6 mm (1/4") bushing (MTS-102 style)	CARGO BAY, DOCK CLAMP

BACKLIT LABELS (LED ESTIMATE AT 60/M)

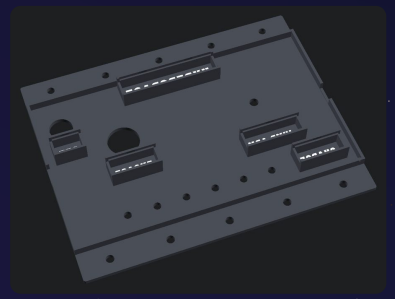
MODE 1 · SCAN PING 2 · RCS 1 · ARM 1 · SCANNER 2 · CARGO BAY 3 · DOCK CLAMP 3 · NAV TGT 2

PANEL OPS_SIDE

Ops side console (hyperspace)



OPS_SIDE · front. Ops side console (hyperspace).



Back: light boxes behind every label.

Nominal size	200 × 160 mm (7.9" × 6.3")
Labels / LEDs	5 labels · 17 LEDs (placeholder map)
Print volume	112 cm³ (body + letters)
3MF check	**watertight, no problems**

WHERE IT GOES

Front pedestal of the reclining cradle, between the kids just ahead of the TD2220 touchscreen, horizontal, top of panel = forward.

CONTROLS

QTY	PART	USED FOR
1	22 mm push button, 22 mm panel hole (industrial 22 mm or mini arcade)	ENGAGE
1	16 mm momentary push button (metal/illuminated ok)	PTT
6	5 mm LED + bezel/holder (or a WS2812B 5 mm through-hole)	jump charge bar (blue)
1	Mini toggle 6 mm + printed missile guard (toggle_guard.3mf)	JUMP ARM

BACKLIT LABELS (LED ESTIMATE AT 60/M)

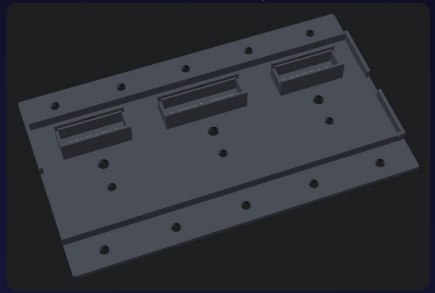
HYPERSPACE 4 · PTT 1 · ENGAGE 2 · JUMP ARM 2 · CHARGE 2

PANEL OVERHEAD_A

Overhead console, left



OVERHEAD_A · front. Overhead console, left.



Back: light boxes behind every label.

Nominal size	200 × 120 mm (7.9" × 4.7")
Labels / LEDs	3 labels · 10 LEDs (placeholder map)
Print volume	84 cm³ (body + letters)
3MF check	**watertight, no problems**

WHERE IT GOES

Pod top band above the monitors, left third (faces the reclined crew, top of panel = up the pod).

CONTROLS

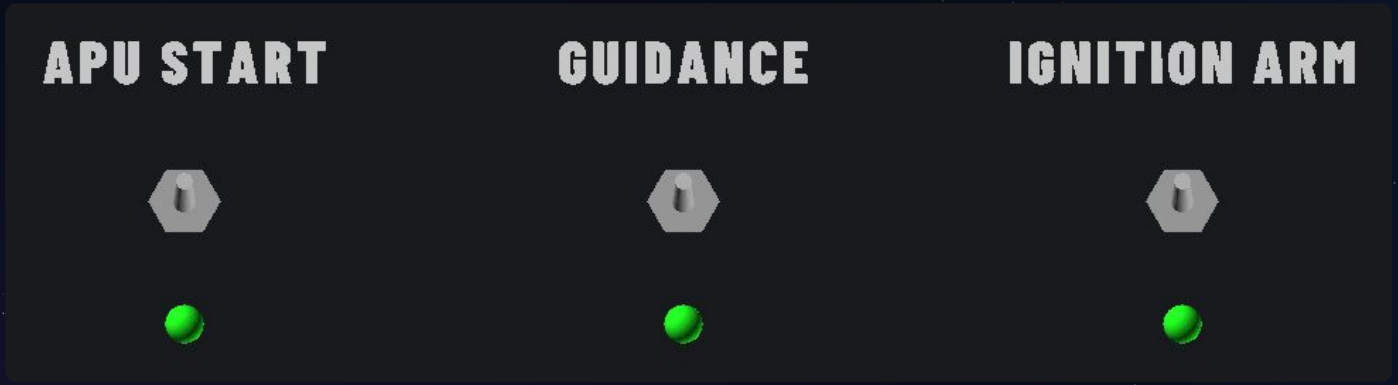
QTY	PART	USED FOR
3	5 mm LED + bezel/holder (or a WS2812B 5 mm through-hole)	AVIONICS status, BATTERY status, FUEL PUMPS status
3	Mini toggle switch SPST/SPDT, 6 mm (1/4") bushing (MTS-102 style)	AVIONICS, BATTERY, FUEL PUMPS

BACKLIT LABELS (LED ESTIMATE AT 60/M)

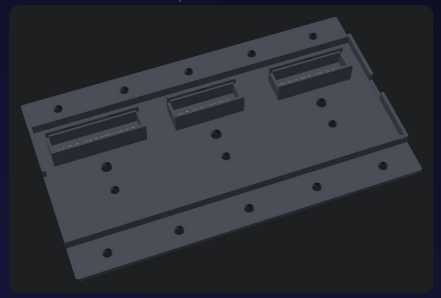
BATTERY 2 · FUEL PUMPS 3 · AVIONICS 2

PANEL OVERHEAD_B

Overhead console, centre



OVERHEAD_B · front. Overhead console, centre.



Back: light boxes behind every label.

Nominal size	200 × 120 mm (7.9" × 4.7")
Labels / LEDs	3 labels · 11 LEDs (placeholder map)
Print volume	85 cm³ (body + letters)
3MF check	**watertight, no problems**

WHERE IT GOES

Pod top band above the monitors, centre third.

CONTROLS

QTY	PART	USED FOR
3	5 mm LED + bezel/holder (or a WS2812B 5 mm through-hole)	APU START status, GUIDANCE status, IGNITION ARM status
3	Mini toggle switch SPST/SPDT, 6 mm (1/4") bushing (MTS-102 style)	APU START, GUIDANCE, IGNITION ARM

BACKLIT LABELS (LED ESTIMATE AT 60/M)

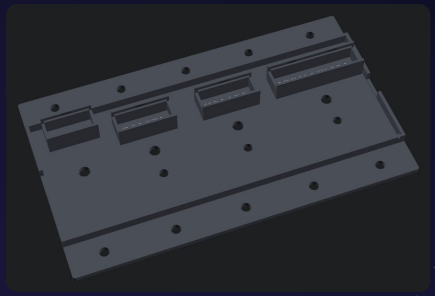
APU START 3 · GUIDANCE 2 · IGNITION ARM 3

PANEL OVERHEAD_C

Overhead console, right



OVERHEAD_C · front. Overhead console, right.



Back: light boxes behind every label.

Nominal size	200 × 120 mm (7.9" × 4.7")
Labels / LEDs	4 labels · 12 LEDs (placeholder map)
Print volume	86 cm³ (body + letters)
3MF check	**watertight, no problems**

WHERE IT GOES

Pod top band above the monitors, right third.

CONTROLS

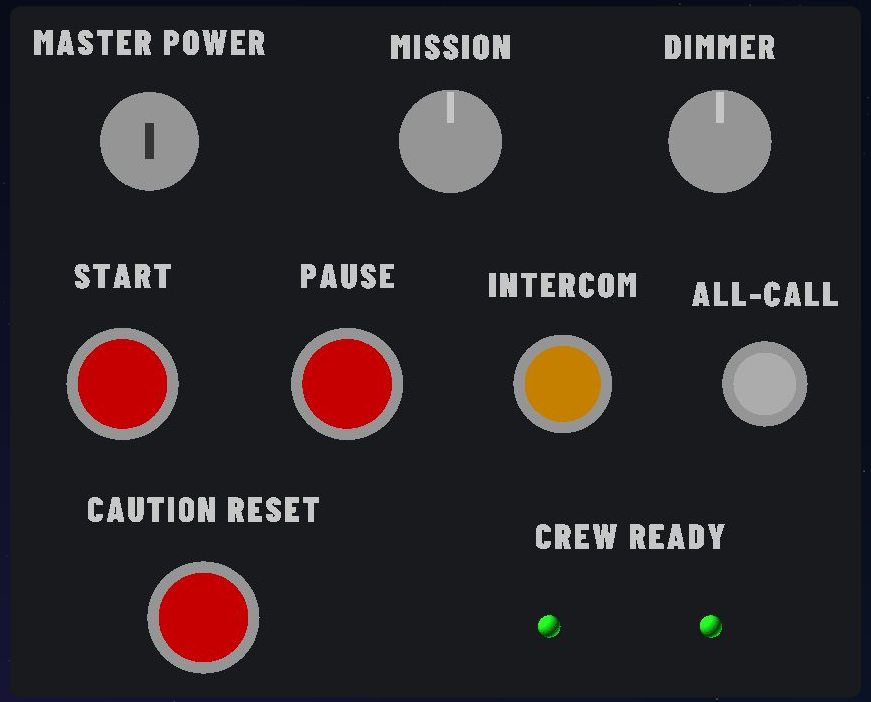
QTY	PART	USED FOR
3	5 mm LED + bezel/holder (or a WS2812B 5 mm through-hole)	BEACON status, CABIN LIGHTS status, STROBE status
3	Mini toggle switch SPST/SPDT, 6 mm (1/4") bushing (MTS-102 style)	BEACON, CABIN LIGHTS, STROBE
1	Mini toggle 6 mm + printed missile guard (toggle_guard.3mf)	EJECT

BACKLIT LABELS (LED ESTIMATE AT 60/M)

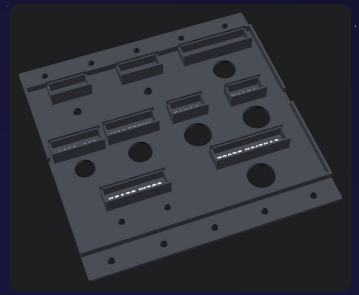
EJECT 2 · STROBE 2 · BEACON 2 · CABIN LIGHTS 3

PANEL CMD_A

Commander console, game controls



CMD_A · front. Commander console, game controls.



Back: light boxes behind every label.

Nominal size	200 × 200 mm (7.9" × 7.9")
Labels / LEDs	9 labels · 23 LEDs (placeholder map)
Print volume	144 cm³ (body + letters)
3MF check	**watertight, no problems**

WHERE IT GOES

Dad's right-hand console on the fixed rear dais.

CONTROLS

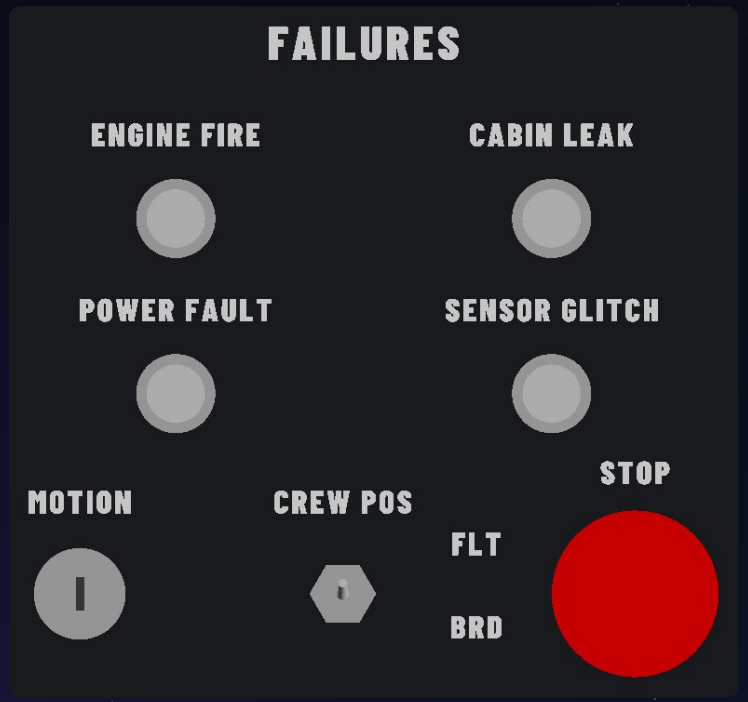
QTY	PART	USED FOR
1	19 mm push button, 19 mm panel hole (metal anti-vandal or mini arcade)	INTERCOM
3	22 mm push button, 22 mm panel hole (industrial 22 mm or mini arcade)	CAUTION RESET, PAUSE, START
1	16 mm momentary push button (metal/ illuminated ok)	ALL-CALL
2	EC11 rotary encoder with push switch + knob	DIMMER, MISSION
1	19 mm key switch, 2 position, key removable in OFF only	MASTER POWER
2	5 mm LED + bezel/holder (or a WS2812B 5 mm through-hole)	OPS ready (green), PILOT ready (green)

BACKLIT LABELS (LED ESTIMATE AT 60/M)

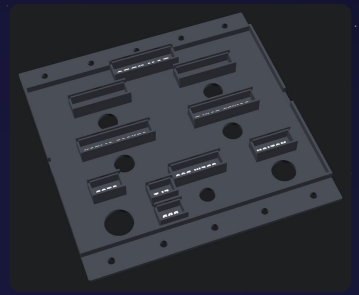
MASTER POWER 3 · MISSION 2 · DIMMER 2 · ALL-CALL 2 · INTERCOM 2 · PAUSE 2 · START 2 · CAUTION RESET 3 · CREW READY 3

PANEL CMD_B

Commander console, failures + crew motion



CMD_B · front. Commander console, failures + crew motion.



Back: light boxes behind every label.

Nominal size	200 × 200 mm (7.9" × 7.9")
Labels / LEDs	10 labels · 22 LEDs (placeholder map)
Print volume	146 cm³ (body + letters)
3MF check	**watertight, no problems**

WHERE IT GOES

Dad's right-hand console on the fixed rear dais, behind CMD_A.

CONTROLS

QTY	PART	USED FOR
4	16 mm momentary push button (metal/ illuminated ok)	CABIN LEAK, ENGINE FIRE, POWER FAULT, SENSOR GLITCH
1	22 mm mushroom E-stop, NC, latching, twist-release	STOP
1	19 mm key switch, 2 position, key removable in OFF only	MOTION
1	Heavy toggle 12 mm bushing, DPDT (ON)-OFF-(ON) momentary, ≥ 15 A @ 12 V DC	CREW POS

BACKLIT LABELS (LED ESTIMATE AT 60/M)

FAILURES 3 · CABIN LEAK 3 · ENGINE FIRE 3 · POWER FAULT 3 · SENSOR GLITCH 3 · STOP 1 · CREW POS 2 · MOTION 2 · FLT 1 · BRD 1

PRINT SETTINGS AND SWITCH TOTALS

	PANELS	LETTERS (SAME 3MF, SECOND OBJECT)	MECHANISMS, MOUNTS, ENCLOSURES
Material	PETG (preferred, tougher, no creep) or PLA; matte black	PETG/PLA **white** (natural/translucent white diffuses best)	PETG (PETG/ASA for the monitor standoffs, PETG for the harness hub)
Nozzle / layer	0.4 mm / 0.2 mm (first layer 0.2)	0.2 mm	0.2 mm
Walls	3 perimeters	2 (boxes and strokes are solid at this size)	4 (monitor standoffs + harness hub 6)
Infill	30 % gyroid (body is solid anyway at 3 mm)	100 %	40 % gyroid (monitor standoffs 50 %, harness hub 40-50 %)
Orientation	**face down**, both objects together (the 3MF is already flipped). Smooth front from the bed; light boxes and the white backing boxes build up	same file, do not move one object	as exported (flat faces down)
Supports	none	none	none (all parts designed support-free)
Bed	smooth PEI or glass gives a glossy face; textured PEI a matte face		

MULTI-MATERIAL AND BED SIZE

Multi-material is the only panel export. Purge/prime tower on, "wipe into infill" off (letters must stay pure white). On a single-nozzle MMU/AMS the colour change is the 3 mm of the face plus the 0.8 mm backing boxes behind it. Every part fits a 220 × 220 mm bed; the overhead, dash and commander panels are split along the grid. Printing on a 250 mm+ bed changes nothing.

SWITCHES FOR ALL PANELS

QTY	PART
2	19 mm push button, 19 mm panel hole (metal anti-vandal or mini arcade)
6	22 mm push button, 22 mm panel hole (industrial 22 mm or mini arcade)
11	16 mm momentary push button (metal/illuminated ok)
4	EC11 rotary encoder with push switch + knob
1	22 mm mushroom E-stop, NC, latching, twist-release
2	19 mm key switch, 2 position, key removable in OFF only
32	5 mm LED + bezel/holder (or a WS2812B 5 mm through-hole)
2	lever slot (printed mechanism below the panel)
1	Heavy toggle 12 mm bushing, DPDT (ON)-OFF-(ON) momentary, ≥ 15 A @ 12 V DC
19	Mini toggle switch SPST/SPDT, 6 mm (1/4") bushing (MTS-102 style)
3	Mini toggle 6 mm + printed missile guard (toggle_guard.3mf)

3MF VALIDATION

22 panel meshes and 31 part files checked (watertight, volume, footprint, triangle counts, lib3mf read-back): **all pass**.

LEVERS, HOUSINGS, MOUNTS AND THE HARNESS HUB

07 OTHER PRINTED PARTS



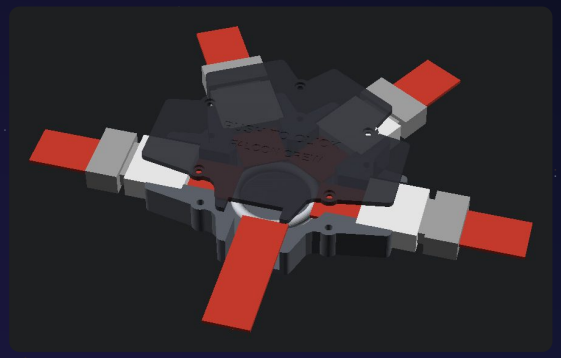
Throttle quadrant. 10 k linear pot, M6 × 30 pivot, red grip; screws into the side-console rails.



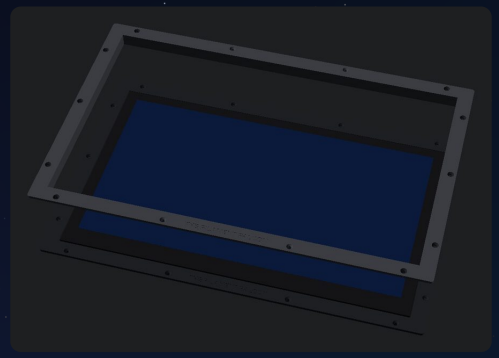
Landing-gear lever. 2 × KW11 micro switches, M4 × 40 pivot, wheel knob.



Joystick housing + adapter. 4 × M5 on the 120 mm grid; radial slots fit bolt circles 44-88 mm.



Harness hub (closed). PETG, 6 perimeters; houses the buckle bodies and the steel load ring.



TD2220 trim ring. 6 half-lapped pieces ≤ 200 mm; never overlaps the optical-touch glass.

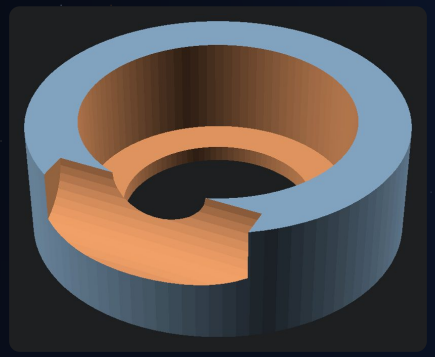


Monitor standoff. VESA 75 + 100 patterns, 4 × M6 through the pod back sheet, plus a steel safety cable.

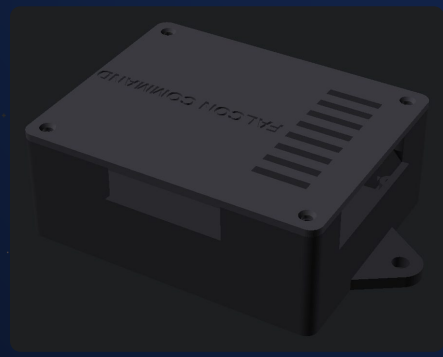
SMALL PARTS



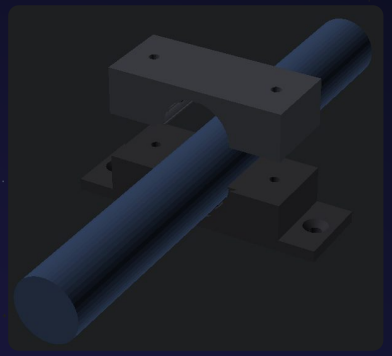
Missile-switch guard. ENG ARM, JUMP ARM, EJECT; hinge pin = 24 mm of 1.75 mm filament.



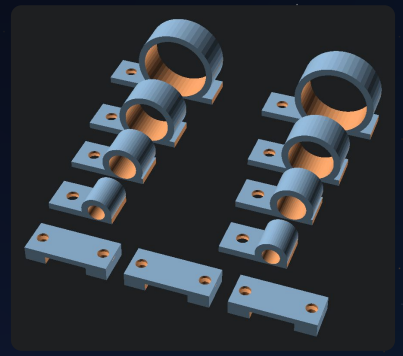
ENGAGE button collar. Raised ring around the 22 mm JUMP ENGAGE button.



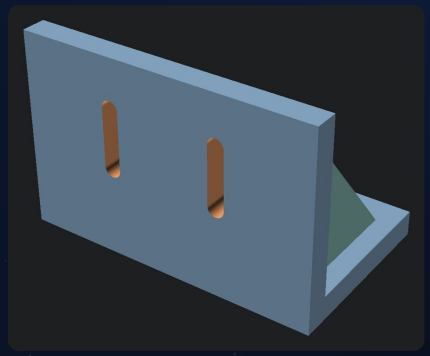
ESP32-S3 enclosure. 70 × 90 mm perfboard carrier, USB-C slot, engraved lid per board.



Pivot cable clamp. Strain relief on both sides of every pivot crossing (30 mm bore).



Cable clips. 6/10/16/24 mm bundle clips + flat LED-strip clip, one #8 screw each.



Limit-switch mount. V-15 roller micro switch, slotted ±10 mm for adjustment.

PRINTED-PARTS LIST from docs/mechanical/README.md

FILE	QTY	COLOUR	NOTES
`toggle_guard_base.3mf` / `toggle_guard_lid_RED.3mf`	3 each	black / red	missile-switch guard for ENG ARM, JUMP ARM, EJECT; hinge = 24 mm of 1.75 mm filament
`engage_button_collar.3mf`	1	black	collar around JUMP ENGAGE
`alignment_pins_x8.3mf`	1 plate	black	seam pins (3 × 16 mm)
`throttle_housing.3mf`, `throttle_lever.3mf`, `throttle_grip.3mf`	1 each	black / black / red	10 k linear pot (16 mm, 6 mm D shaft), M6 × 30 pivot; housing screws into the side-console rails
`gear_lever_box.3mf`, `gear_lever_arm.3mf`, `gear_lever_wheel_knob.3mf`	1 each	black / grey / white	2 x KW11 micro switches, M4 × 40 pivot, hangs on 4 x M3 × 12 countersunk
`joystick_housing.3mf`, `joystick_adapter_plate.3mf`	2 each	black / grey	4 x M5 on the grid (120 mm); adapter has 8 radial slots r 22-44 mm + optional square pattern (`SQ_PITCH`)
`td2220_trim_corner_front_left/right`, `td2220_trim_corner_rear_left/right`, `td2220_trim_middle_front/rear.3mf`	1 each	black	TD2220 trim ring (558.5 × 357.4 outer) in 6 half-lapped pieces ≤ 200 mm; never overlaps the screen (optical touch)
`td2220_vesa100_riser.3mf`	1	black	VESA 100 riser for the TD2220 on the console shelf (`TILT` param)
`monitor_standoff_C275B_vesa100.3mf`, `monitor_standoff_C275W_vesa75.3mf`	1 each	black	pod monitor standoffs (VESA 75 + 100 patterns), 4 x M6 through the back sheet. **PETG/ASA, 6 walls, 50 % gyroid, plus a steel safety cable** (the pod parks the monitors face-down)
`harness_hub_base.3mf`, `harness_hub_lid.3mf`	2 each	black	4-point harness hub for the buckle kit (see below)
`pivot_cable_clamp_base.3mf`, `pivot_cable_clamp_cap.3mf`	8 each	black	strain relief on both sides of every pivot crossing (`BORE` param, default 30 mm)
`esp32_enclosure_base.3mf`	3	black	70 × 90 perfboard carrier for an ESP32-S3-DevKitC-1, USB-C slot, M5 tabs 120 mm apart
`esp32_enclosure_lid_PILOT/OPS/COMMAND.3mf`	1 each	black	engraved name
`cable_clips_set.3mf`	3 plates	black	6/10/16/24 mm bundle clips + flat LED-strip clip, one #8 screw each
`limit_switch_mount.3mf`	11	black	V-15 roller micro switch, slotted ±10 mm (8 cradle/pod limits + lock/lever switches, 3 READ-only position senses)

Not printed: webbing, buckles, the steel load ring, anchors (forged eye bolts), actuator brackets, shafts, bearings, pawls, stop pins (see `frame/BOM.md`).

FILAMENT ESTIMATE (BOM/TOOLS)

Black ≈ **5.26 kg** (3601.0 cm³) and white ≈ **0.06 kg** of PETG, incl. 15 % for purge, brims and failed prints. The BOM buys 6 × 1 kg black and 1 × 1 kg white.

TWO CURVED WINDOW MONITORS, A TOUCHSCREEN AND THE COMMANDER MONITOR

08 DISPLAYS AND CABLING

	SCEPTRE C275B-144R	SCEPTRE C275W-1920RN	VIEWSONIC TD2220
Use	PILOT window (left kid), in the pod	OPS window (right kid), in the pod	centre console touchscreen, in the cradle
Panel	27" 1920 × 1080, up to 144 Hz	27" 1920 × 1080, 75 Hz	21.5" 1920 × 1080, 60 Hz, optical 2-point touch (USB-B)
Size W x H x D (no stand)	618 × 363.5 × 62.5 mm	616.5 × 366.5 × 65 mm	511.5 × 310.4 × 66.5 mm
Mass (no stand)	3.80 kg	3.40 kg	4.0 kg
Curve	1800R	1800R	flat
VESA	100 × 100, M4	**75 × 75**, M4	100 × 100, M4
Video inputs	HDMI, DisplayPort, DVI	**HDMI + VGA only**	**VGA + DVI-D only** (no HDMI/DP)
Active area	~598 × 336	~598 × 336	~476 × 268

MONITOR MOUNTS (POD)

`cad/parts/monitor_mount.scad` → `monitor_standoff_C275B_vesa100.3mf` and `monitor_standoff_C275W_vesa75.3mf`: one printed hat-channel standoff per monitor, bolted through the pod's 12 mm back sheet with 4 x M6 carriage bolts. The plate has both VESA 75 and VESA 100 hole patterns; only the height differs (64.0 / 61.4 mm) so both screen centres sit exactly 600 mm from the pod pivot (~540 mm from the 12 yr's eyes in flight). Screws: M4 × 20 (C275B), M4 × 16 (C275W) - check the insert depth first, do not bottom out.

PARKED FACE-DOWN

The pod parks the monitors face-down above the kids (pod UP = 95°), so each monitor gets a 2 mm steel safety cable from a VESA screw to an eye screw in the back sheet (≤ 50 mm slack). PETG/ASA, 6 perimeters, 50 % gyroid. A steel VESA plate on 64 mm spacers is an equal substitute.

TD2220 CENTRE CONSOLE

The TD2220 lies (near) flat in the cradle's centre console between the kids (console x = ±280 mm, seat centres ±460 mm), on a printed VESA 100 riser (`td2220_vesa100_riser.3mf`, 190 × 140 × 40 mm, `TILT` parameter, default 0) screwed to a plywood shelf inside the console. Because the console reclines with the cradle, "flat" in the cradle ends up tilted 35° toward the reclined kids in flight - the screen faces them (screen normal ~55° from horizontal).

GPU OUTPUTS AND CABLE ROUTING

RX 6800 XT: 2 × DP, HDMI, USB-C

The reference RX 6800 XT has **2 x DisplayPort 1.4, 1 x HDMI 2.1, 1 x USB-C (DP alt mode)** and drives at most **4 displays at once**.

GPU PORT	CABLE	DISPLAY	MODE
DP 1	DP 1.4 cable, 3-5 m	C275B-144R (PILOT) DP in	1920 × 1080 @ 144 Hz
HDMI 2.1	HDMI cable, 3-5 m	C275W-1920RN (OPS) HDMI in (its only digital input)	1920 × 1080 @ 75 Hz
DP 2	**DP → DVI-D single-link** adapter or cable (passive is fine: DP++)	TD2220 DVI-D in	1920 × 1080 @ 60 Hz
USB-A (PC)	USB-A → USB-B, ≤ 5 m (or active extension)	TD2220 touch	HID touch
USB-C	USB-C → DP/HDMI	4th display: the COMMANDER monitor (31.5" 1080p at dad's seat)	1920 × 1080 @ 60 Hz

All four heads are now in use. The actual cabling may differ from this plan. The game assigns screen roles by xrandr output name and physical size, not by port. On the owner's rig (v0.3) the outputs are DisplayPort-2 = left window, DisplayPort-1 = right window, DisplayPort-0 = TD2220 and HDMI-A-0 = 31.5" Commander (see `docs/game/README.md`, "The four screens").

CABLES ACROSS THE PIVOTS

Routing: the two monitor cables + power cross the **pod pivot**, the TD2220 DVI + USB + power cross the **cradle pivot**. Each bundle gets 2 printed `pivot_cable_clamp`s (one on the fixed side, one on the moving side, ≤ 60 mm from the axis) and a 200 mm U service loop in braided sleeve. Use cables with molded strain relief and screw-lock (DVI) or latching (DP) connectors.

TRIM RING AND TOUCH

The **trim ring** (outer 558.5 × 357.4 mm) closes the gap between the screen edge and the plywood cut-out and **never overlaps the screen**: the TD2220's optical touch sensors look across the glass from the bezel, so anything raised over the bezel would block touches. The ring top sits 1 mm below the glass. Six pieces, all ≤ 220 mm: 4 corners `td2220_trim_corner_*` (200 × 179 / 195 mm) and 2 straight middles `td2220_trim_middle_*` (190.5 mm incl. laps), half-lapped, one countersunk M4 wood screw through every lap. Ports are on the rear edge (toward dad).

FIT BETWEEN THE KIDS

Fit between the kids' seats: trim 558.5 mm wide in a console 560 mm wide; seat boxes are 340 mm wide at ±460 → 290 mm inner edges → 10 mm side clearance between console and seat boxes. Kids' hands reach it reclined (it rides with them).

THREE ESP32-S3 USB JOYSTICKS

09 ELECTRONICS

BOARD	USB NAME	VID:PID	WHERE	INPUTS	EXTRA
ESP32-S3-DevKitC-1 #1	**Falcon Pilot**	1209:0003	left kid, PILOT	3-axis stick + trigger, throttle pot, 2 × 74HC165 (12 used, incl. PTT), 1 EC11 (FLIGHT ASSIST)	
ESP32-S3-DevKitC-1 #2	**Falcon Ops**	1209:0004	right kid, OPS	3-axis stick + trigger, 2 × 74HC165 (14 used, incl. PTT), 1 EC11 (OPS MODE)	
ESP32-S3-DevKitC-1 #3	**Falcon Command**	1209:0005	dad + overhead console	4 × 74HC165 (22 used, incl. INTERCOM PTT + ALL-CALL), 2 EC11 (MISSION, DIMMER knob), 3 read-only crew-position optos (GPIO39/40/41)	USB CDC serial, drives the ONE WS2812B data line

CONTROLS AS THE CAD PANELS DEFINE THEM

The control set matches the CAD panel layouts (`cad/tools/layouts.py`, `shared/panel_labels.json`): flight-assist mode, ops mode and mission select are EC11 encoders with push (the game keeps the selected mode and lights the matching label), the dimmer is an EC11 "virtual knob" (firmware keeps a 0-255 level, reports it as an axis and dims the strip), NAV TGT is a push button, FLAPS is an ON-OFF-ON toggle.

WHY THE ESP32-S3?

it has a native USB port, so it enumerates as a genuine HID joystick and (for Falcon Command) also a serial port at the same time. The ESP8266 and the original ESP32 have no native USB and would need a USB-serial bridge plus PC-side software. The S3 also has 10 ADC1 pins, enough RAM and 240 MHz dual core, more than enough to poll 32 switches at 1 kHz and render a 200+ LED strip at 60 fps.

ONE FIRMWARE, THREE BOARDS

PlatformIO + Arduino + TinyUSB. Each board is a USB HID joystick with 8 × 16-bit axes and 64 buttons; Falcon Command is also a USB CDC serial port that drives the single WS2812B strip. Which switch is which button lives in `firmware/include/config_*.h`; `shared/input_map.json` and the 02 wiring diagrams are generated from those headers.

SWITCH INPUTS

Switches go through daisy-chained 74HC165 shift registers (3 Pilot, 3 Ops, 4 Command), analog axes to ADC1 pins at 3.3 V. The last chip's SER pin is tied to 3V3, so the firmware can tell when no chain is connected and falls back to joystick-only mode.

ELECTRONICS PARTS

from docs/electronics/README.md (counts include spares)

CONTROLLERS AND LOGIC

QTY	PART	NOTES
3	ESP32-S3-DevKitC-1 (N8 or N8R8)	must be the S3 with the "USB" + "UART" ports
3	USB-C data cables (1-2 m)	from each board's **USB** port to the PC; a powered USB 3 hub is fine
10 (+2 spare)	74HC165 DIP-16 + sockets	3 Pilot, 3 Ops, 4 Command
12	10 kΩ bussed resistor network, 9-pin SIP (e.g. 4609X-101-103)	one per 74HC165 (8 pull-ups); or 96 × 10 kΩ resistors
20	100 nF ceramic capacitors	one per chip + spares
5	10 µF electrolytic/ceramic	at each board and at the far end of the overhead cable
1	74AHCT125 DIP-14 (or SN74AHCT1G125)	3.3 V → 5 V LED data
2	330 Ω resistors	LED data series resistor (+1 spare)
2	100 Ω resistors	series on CLK for the long overhead cable
8	10 kΩ resistors	3 opto pull-ups (cradle reclined, pod down-locked, pod up-locked), spares
3	perfboard / protoboard 7 × 9 cm, pin headers, Dupont + JST-XH connectors	one carrier per DevKitC-1
1	CAT5/CAT6 patch cable	Falcon Command → overhead console chain (CLK, LOAD, DATA, 3V3, GND pairs)

PANEL CONTROLS

QTY	PART	USED FOR
22	6 mm mini toggle ON-OFF (SPST/SPDT)	Pilot AIRBRAKE, NAV LIGHTS; Ops 5 power buses, CARGO BAY, DOCK CLAMP; overhead 9 checklist/ lights
2	6 mm mini toggle ON-OFF-ON (SPDT centre off)	Pilot FLAPS (+1 spare)
4	6 mm mini toggle for printed guards	ENG ARM, JUMP ARM, EJECT (+1)
8	22 mm arcade momentary (colours per CAD)	IGNITION, BOOST, ENGAGE, START, PAUSE, CAUTION RESET (+2)
3	19 mm momentary	FIRE SUPPR, INTERCOM (+1)
10	16 mm momentary	HORN, VIEW, SCAN PING, NAV TGT, 4 failures (+2)
3	16 mm momentary (intercom, on the CAD panels since v0.3)	PILOT PTT (PILOT_SIDE), OPS PTT (OPS_SIDE), ALL-CALL / PA (CMD_A)
3	USB headsets with microphone (any brand; a serial number helps tell identical ones apart)	intercom, on the PC (not the boards)
5	EC11 rotary encoder with push + knob	FLIGHT ASSIST, OPS MODE, MISSION, DIMMER (+1)
1	10 kΩ linear slide or rotary pot (B10K)	Pilot throttle lever
1	lever microswitch	landing gear lever (in the printed gear box)
1	19 mm key switch, 2 position	MASTER POWER (game input)
2	your 3-axis joysticks with top button	Pilot, Ops

FIRST TEST: ONE JOYSTICK, ONE BOARD

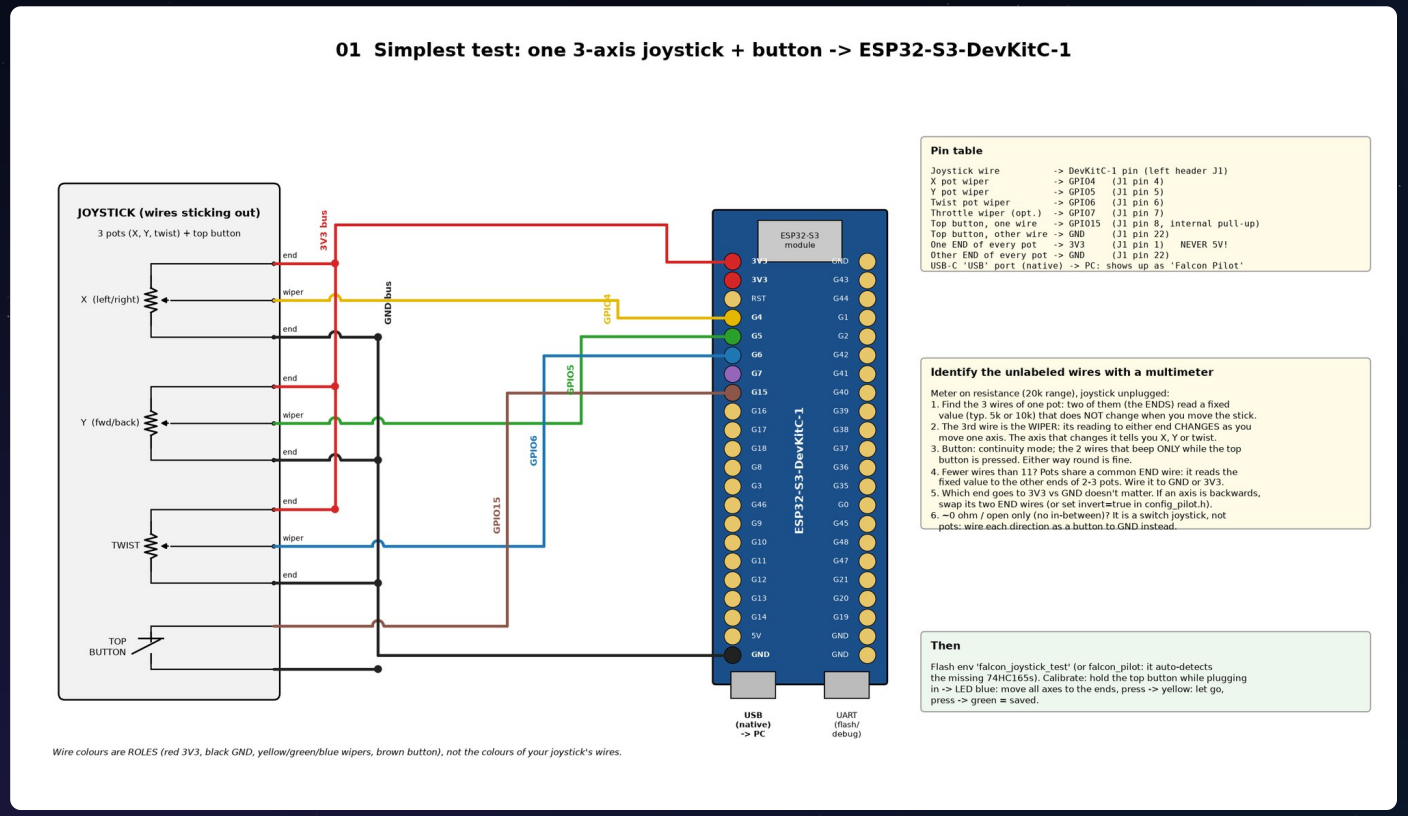


Diagram 01 · start here. One 3-axis joystick + button straight to a DevKitC-1, and how to identify unlabeled wires with a multimeter.

JOYSTICK WIRE	DEVKITC-1 PIN
X pot wiper	GPIO4
Y pot wiper	GPIO5
Twist pot wiper	GPIO6
Throttle pot wiper (optional)	GPIO7
Top button (one side)	GPIO15 (internal pull-up)
Top button (other side)	GND
One end of each pot	3V3 (**never 5 V**)
Other end of each pot	GND

CALIBRATE

(stored in flash, survives reflashing the same board): hold the top button (or the board's BOOT button) while plugging in, keep holding ~1 s until the on-board LED turns **blue** → move every axis to both ends → press → **yellow**: let go of the stick → press → **green** x3 = saved. Without calibration the firmware assumes full pot travel and takes the resting position at power-up as centre.

FALCON PILOT WIRING

generated from the firmware config headers

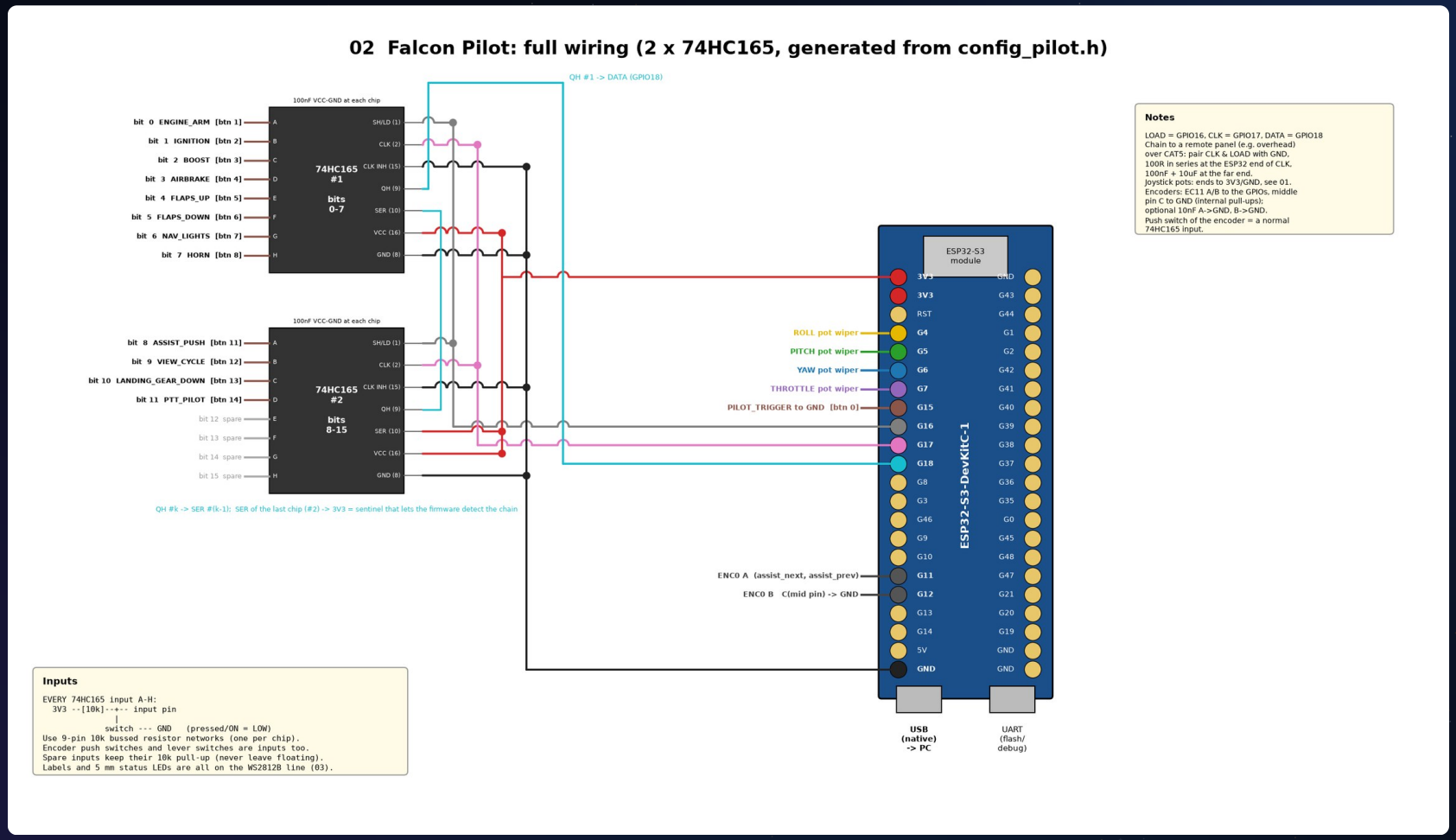


Diagram 02 · Falcon Pilot wiring. 74HC165 chain with every switch named, joystick, throttle pot and the FLIGHT ASSIST encoder.

FALCON OPS WIRING

generated from the firmware config headers

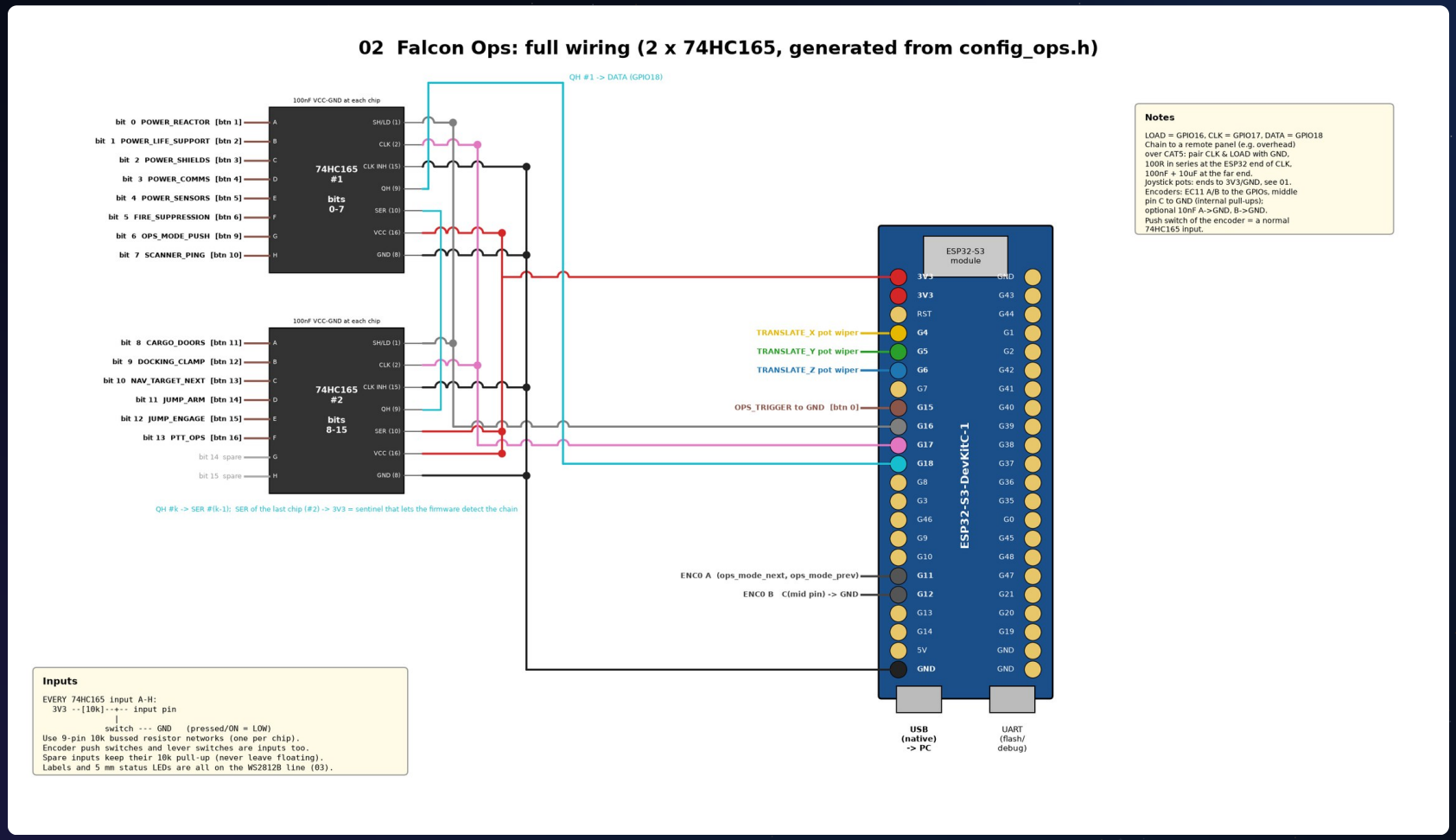


Diagram 02 · Falcon Ops wiring. 74HC165 chain for power toggles, payload and jump controls, RCS joystick and OPS MODE encoder.

FALCON COMMAND WIRING

generated from the firmware config headers

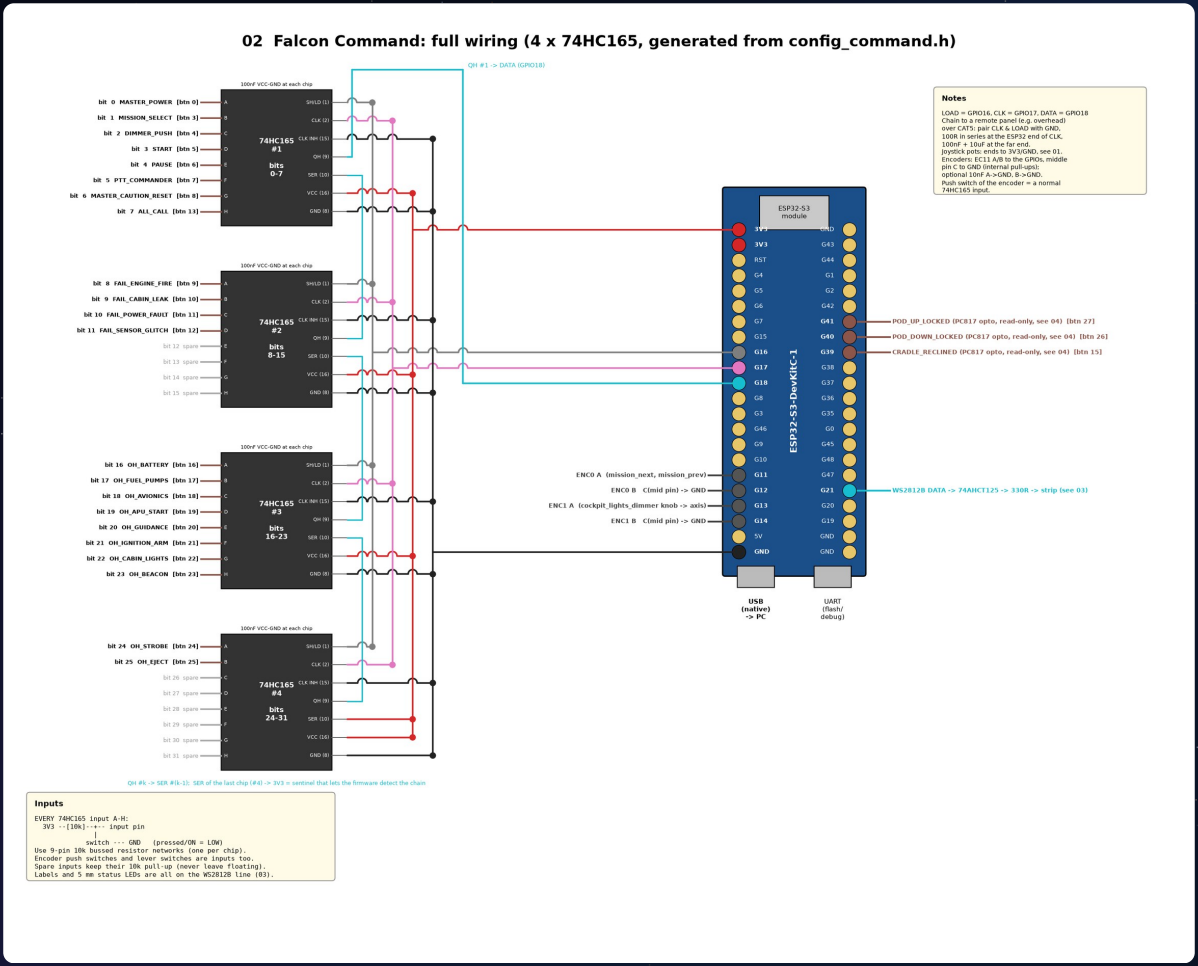


Diagram 02 · Falcon Command wiring. Dad's panel + overhead chips (over a CAT5/6 cable), MISSION and DIMMER encoders, LED data line and the three read-only crew-position senses (GPIO39/40/41).

LED STRIP POWER

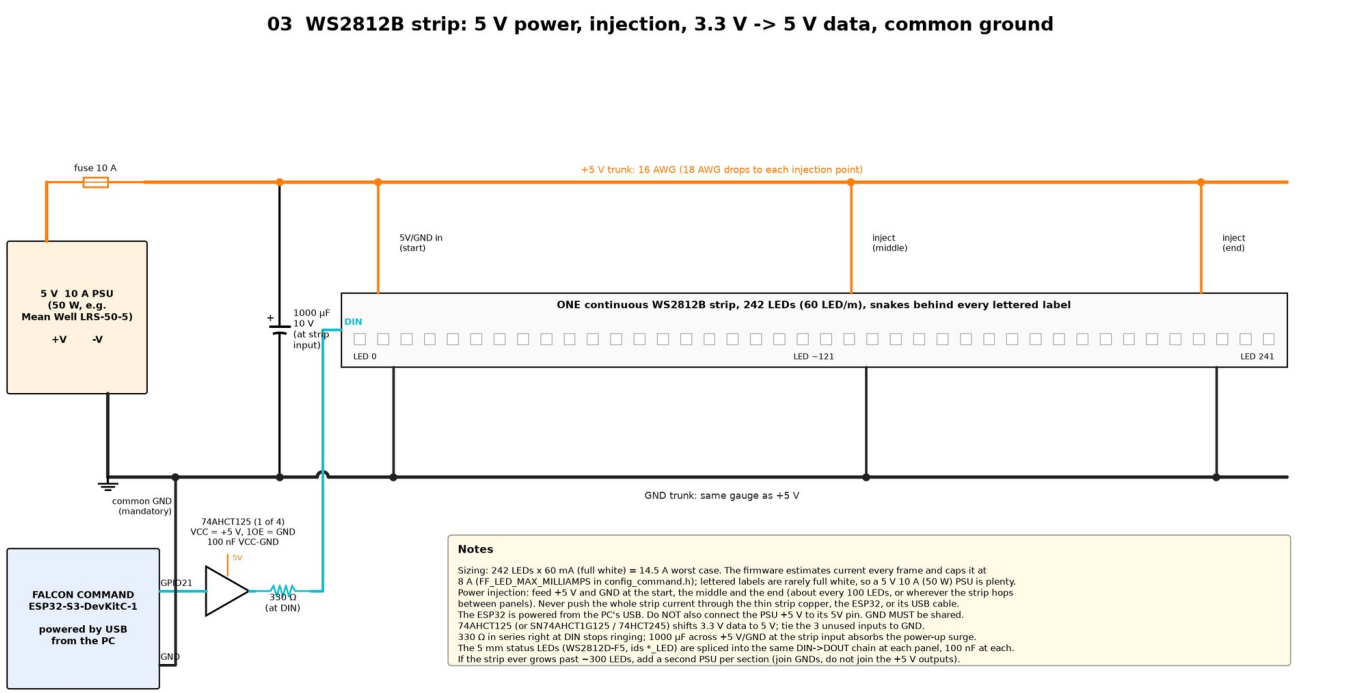


Diagram 03 · LED strip power: 5 V supply sizing, power injection at start/middle/end, 1000 µF at the strip input, 330 Ω data resistor, 74AHCT125 level shifter, common ground.

QTY	PART	NOTES
1	WS2812B strip 5 V, 60 LED/m, 5 m reel (300 LEDs)	~140 LEDs behind labels + ~60 lost in panel hops; 10 mm wide for the printed channels
40	5 mm through-hole addressable LEDs, WS2812D-F5 (GRB)	the 32 status/indicator LED holes in the CAD panels, spliced into the same data line (`*_LED` ids)
1	5 V 10 A (50 W) PSU, e.g. Mean Well LRS-50-5	242 LEDs x 60 mA = 14.5 A absolute worst case; firmware caps at 8 A
1	1000 µF 10 V electrolytic	across 5 V/GND at the strip input
32	100 nF ceramic	one at each 5 mm WS2812D-F5 (VDD-GND)
1	10 A fuse + holder (blade or 5x20)	PSU output
-	16 AWG red/black (trunk), 18 AWG (injection drops), 3-pin JST-SM pigtails	power injection at start, middle, end

PIN PLAN AND FIRMWARE BEHAVIOUR

FUNCTION	GPIO	NOTES
Stick X / Y / twist	4 / 5 / 6	ADC1, pot ends on 3V3/GND
Throttle lever pot (Pilot)	7	ADC1
Spare analog	1, 2	ADC1
Stick top button	15	to GND, internal pull-up; also the calibration button
74HC165 SH/LD (load)	16	all chips
74HC165 CLK	17	all chips
74HC165 QH data in	18	from chip #1; chip #k QH → chip #(k-1) SER; last SER → 3V3
EC11 encoder 0 A / B	11 / 12	C pin to GND
EC11 encoder 1 A / B	13 / 14	Command: DIMMER knob
WS2812B data (Command)	21	via 74AHCT125 + 330 Ω
CRADLE_RECLINED sense (Command)	39	PC817 collector, active LOW, HID button 15
POD_DOWN_LOCKED sense (Command)	40	PC817 collector, active LOW, HID button 26
POD_UP_LOCKED sense (Command, optional)	41	PC817 collector, active LOW, HID button 27; leave open if not fitted
On-board RGB status LED	48 (v1.0) / 38 (v1.1)	see `PIN_STATUS_RGB`

Avoided: GPIO0/3/45/46 (strapping), 19/20 (USB), 26-37 (flash/PSRAM), 43/44 (UART log). GPIO39-41 are the JTAG pins; they are free because the boards use the native USB port (no external JTAG).

FIRMWARE BEHAVIOUR

- **Axes**: ADC1, 11 dB attenuation, 16x oversampling, EMA smoothing, per-axis deadzone, edge snap. BIPOLAR axes (stick) map min/centre/max to -32767/0/+32767; UNIPOLAR (throttle, dimmer) map min/max to -32767/+32767. Axes that never moved during calibration report 0 ("not wired").
- **Buttons**: 74HC165 chain read at loop rate (~1-2 kHz), 6 ms debounce. Toggles are reported as held buttons while ON; each encoder click becomes a 35 ms button pulse (queued, so fast spins are not lost). An encoder whose `FF_ENC` has an axis (Command's DIMMER) is a virtual knob instead: a 0-255 level, 8 per click, saved to flash 3 s after the last click, reported on that HID axis.
- **Fallback**: the last 74HC165's SER pin is tied to 3V3. If the firmware does not see those 1s after the chain, there is no chain, and it ignores shift-register buttons (joystick-only mode). It re-checks continuously, so you can wire the chain later without reflashing. Status LED: green = chain found, blue = fallback.
- **Report rate**: on change up to 1 kHz, plus a 100 ms keep-alive.
- **Calibration**: see docs/electronics/README.md.

LINUX MINT AND WINDOWS SETUP

from docs/electronics/linux_setup.md

1 · UDEV RULE (ONCE)

```
sudo cp firmware/udev/99-filament-falcon.rules /etc/udev/
rules.d/
sudo udevadm control --reload && sudo udevadm trigger
# unplug + replug the three boards
ls -l /dev/falcon-command /dev/serial/by-id/      #
Falcon Command's CDC serial port
```

2 · IS IT ENUMERATING?

```
lsusb | grep 1209
#   ID 1209:0003 Filament Falcon Falcon Pilot
#   ID 1209:0004 Filament Falcon Falcon Ops
#   ID 1209:0005 Filament Falcon Falcon Command
dmesg -w            # plug a board in: "input: Filament
Falcon Falcon Pilot as /devices/..." and for Command also
"cdc_acm ... ttyACM0"
cat /proc/bus/input/devices | grep -A4 Falcon
```

3 · TEST THE JOYSTICK

```
sudo apt install evtest joystick jstest-gtk
evtest                        # pick "Filament Falcon
Falcon Pilot"; move the stick -> ABS_X / ABS_Y / ABS_RZ,
                              # throttle -> ABS_THROTTLE,
top button -> BTN_TRIGGER (button 0)
jstest /dev/input/js0         # numbers: axis 0..7, button
0..63
jstest-gtk                    # GUI: shows all three Falcon
devices with live bars
```

4 · TEST THE LEDS

```
sudo apt install picocom
picocom -b 115200 --omap crlf --echo /dev/falcon-command
ID?
ALL 0 0 255 PULSE
FIND 17
MAP?
```

FLASHING

- Easiest: plug the **UART** port and run `cd firmware && pio run -e falcon_pilot -t upload`.
- Native **USB** port only: hold **BOOT**, tap **RESET**, release BOOT (board enters ROM download mode and shows up as `/dev/ttyACM*`), run the upload, then tap RESET.
- Joystick-only test: `pio run -e falcon_joystick_test -t upload`.

WINDOWS

1. Plug in the board's **USB** port. No drivers needed (HID + the in-box `usbser` CDC driver).
2. `Win+R` → `joy.cpl` → "Falcon Pilot" / "Falcon Ops" / "Falcon Command" → Properties. The test page shows X/Y axes, Z rotation (twist), Slider (throttle/dimmer) and buttons. joy.cpl only displays buttons 1-32; buttons 33-64 still work in games (DirectInput/SDL).
3. Device Manager → Ports: "USB Serial Device (COMx)" is Falcon Command's LED port.
4. If you change the HID descriptor but keep the PID, Windows may cache the old layout: uninstall the device in Device Manager (View → Show hidden devices) and replug.

EVERY LABEL GLOWS, AND THE GAME PICKS THE COLOUR

10 LED LIGHTING SYSTEM

242
LEDs on the strip (WS2812B, GRB, 60/m)

95
named segments: backlit labels + 5 mm status LEDs

25
5 mm WS2812D-F5 status LEDs spliced into the same line

172
LEDs assigned; the rest are hops between panels

ONE DATA LINE, MANY PANELS

Falcon Command drives the whole strip from GPIO21 through a 74AHCT125 level shifter. The game never talks to LEDs directly: it sends short text lines over USB serial (through a small Python bridge, because Godot has no serial API) and the firmware renders at 60 fps with global brightness, the DIMMER knob and a current limit (capped at 8 A) on top.

PLACEHOLDER MAP

PLACEHOLDER map until the real strip is installed and measured. label id → first LED index (0 = first LED after Falcon Command) + LED count. Labels use est. LEDs at 60/m from CAD; *_LED entries are 5 mm through-hole WS2812D-F5 spliced into the same data line. Edit freely, then rebuild/flash Falcon Command (the firmware compiles this file in). Label ids from: shared/panel_labels.json (CAD) + 5 mm LED holes from cad/tools/layouts.py.

SEGMENTS PER PANEL, IN STRIP ORDER

PANEL	LABELS	STATUS LEDS	LEDS
CMD_A	9	2	23
CMD_B	10	0	22
PILOT_SIDE	7	0	8
OPS_SIDE	5	1	17
OPS_DASH_A	6	5	19
OPS_DASH_B	8	2	17
OVERHEAD_C	4	3	12
OVERHEAD_B	3	3	11
OVERHEAD_A	3	3	10
PILOT_DASH_A	7	4	18
PILOT_DASH_B	8	2	15
Total	**70**	**25**	**172**

BOOT AND IDLE

1. Power-up: cyan comet sweeps from LED 0 to the end of the strip (1.4 s), then every label flashes white and fades into its current state (1 s).
2. Until the game sends anything: **idle glow** (all labels warm white, dim) so the panels are readable.
3. Game closes the port: idle glow again.

THE SERIAL LED PROTOCOL

game → Falcon Command, shared/led_protocol.md

COMMAND	EFFECT	REPLY
`SEG <LABEL_ID> <r> <g> <b> [MODE]`	set every LED of a label (ids in `shared/led_segments.json`: the CAD label ids from `shared/panel_labels.json`, plus `<ID>_LED` for the 5 mm status LEDs)	none / `OK`
`ALL <r> <g> <b> [MODE]`	set the whole strip	none / `OK`
`IDX <start> <count> <r> <g> <b> [MODE]`	set raw LED indices (for mapping / tests)	none / `OK`
`CLEAR`	all LEDs off	none / `OK`
`IDLE`	back to the idle glow (all labels warm white, dim)	none / `OK`
`BRIGHT <0-255>`	global brightness (default 96)	none / `OK`
`ANIM BOOT` / `ANIM ALERT` / `ANIM STOP`	play the boot sweep / 3 red flashes over everything, or stop	none / `OK`
`FIND <index>`	light ONLY LED `<index>` white (use it to measure the real segment map)	`FIND <index>`
`MAP?`	dump the compiled-in segment map	`MAP <ID> <start> <count>` per label, then `END <segments> <leds>`
`ID?`	identify	`ID FALCON_COMMAND <fw-version> LEDS <n> SEGMENTS <m>`
`STATE?`	seat sensor + brightness	`STATE SEAT UP|DOWN BRIGHT <b> DIMMER <0-255>`
`PING`	liveness	`PONG`
`ACK ON` / `ACK OFF`	per-command `OK` replies on/off	`OK`

MODES

MODE	LOOK	SUGGESTED USE
`SOLID` (default)	steady	normal "on" / selected
`BLINK`	1 Hz, 50 %	waiting for this input (checklist step)
`FAST`	4 Hz	warnings, MASTER CAUTION, fire
`PULSE`	0.5 Hz breathe	armed / standby (JUMP ARM, ENGINE ARM)
`OFF`	dark	label not available

EXAMPLE SESSION

```
> ID?
< ID FALCON_COMMAND 0.1.0 LEDS 238 SEGMENTS 92
> BRIGHT 120
> CLEAR
> SEG BATTERY 0 255 0
> SEG BATTERY_LED 0 255 0
> SEG FUEL_PUMPS 255 160 0 BLINK
> SEG ENG_ARM 255 0 0 PULSE
> SEG MASTER_CAUTION_LED 255 0 0 FAST
> SEG NOT_A_LABEL 1 2 3
< ERR unknown label
< EVT SEAT UP
```

FRAMING

- ASCII, one command per line, terminated by `\n` (`\r\n` also accepted). Max 127 characters per line.
- Keywords and label ids are case-insensitive. Fields separated by single spaces.
- Colours are decimal `0..255`. The firmware applies global brightness, the dimmer pot and a current limit on top.
- Changes are rendered at 60 fps; there is no "show" command. Sending the same state again is harmless.

11 THE GAME

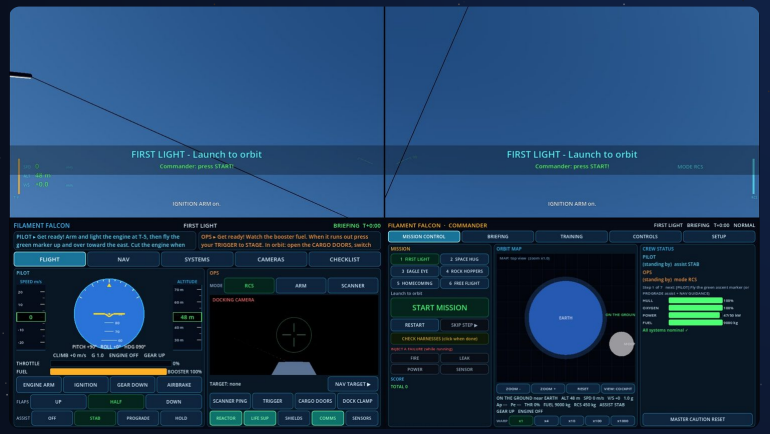
A two-seat (Pilot + Ops) spaceship cockpit game for Godot 4. Everything (planet, Moon, stars, ship, station, sounds) is generated procedurally at start-up, so the repo stays small and has no downloaded assets (see CREDITS.md).

ENGINE

Godot 4 (README: 4.7.x stable, standard build), Forward+ renderer on Vulkan (Mesa RADV on the RX 6800 XT), GDScript. `godot --path game` opens cockpit mode (one fullscreen window per screen) when it finds ≥ 3 screens, otherwise a tiled dev mode. Four windows: left and right out-the-window views, the kids' touch console and the Commander screen (with three screens the Commander becomes a console tab).

GRAPHICS NOTES

- **Window views stay clean.** Only a small, semi-transparent HUD is drawn on the windows: speed/altitude/vertical speed bottom-left, throttle and fuel bars, flight-path markers, a faint horizon line with a gap in the middle, and short cues (countdown, sink rate, AoA, g, hull). The full instruments (attitude ball, speed and altitude tapes, orbit, Ap/Pe, assist, gear, flaps, engine) are on the kids' console, and a telemetry summary is on the Commander screen.
- **Exposure** is scripted, not automatic, so bots and screenshots are repeatable: the bright daytime sky stops down (about 0.78), space sits at about 0.92, and night inside the atmosphere opens up. ACES tonemapping.
- **Everything is procedural**, including the "textures": planet (continents, oceans with sun glint and Fresnel, cloud deck with cloud shadows and clear



Dev mode: both window views, the kids' console and the Commander screen tiled in one window.

ROLE	WHAT IT SHOWS	HARDWARE ON YVES'S RIG	XRANDR OUTPUT
LEFT WINDOW	out-the-window view + minimal HUD	Sceptre C275B-144R, 27" curved, 1920x1080 @ up to 144 Hz	DisplayPort-2
RIGHT WINDOW	out-the-window view + minimal HUD	Sceptre C275W-1920RN, 27" curved, 1920x1080 @ 75 Hz	DisplayPort-1
KIDS' CONSOLE	the crew's instruments and controls (touch)	ViewSonic TD2220, 22" optical touch, 1920x1080 @ 60 Hz	DisplayPort-0
COMMANDER	mission control for the grown-up (mouse + keyboard)	31.5" 1920x1080 monitor	HDMI-A-0

At start-up the game reads `xrandr --listmonitors` (output names and physical sizes) and assigns roles in this order: The defaults are `console_output="DisplayPort-0"`, with everything else automatic. On Yves's rig that gives LEFT = DisplayPort-2, RIGHT = DisplayPort-1, CONSOLE = DisplayPort-0 and COMMANDER = HDMI-A-0. The start-up log prints the result (`display: cockpit layout on 4 screens …`). To change it, open **Commander → SETUP → SCREENS**: - ◀ ▶ cycles each role through AUTO, screen 0, 1, 2, ...

weather over the pad and runway, atmosphere scattering and limb), terrain (farm plots, roads, forest, rock; craters on the Moon), and the PBR panel shader used on the ship and station (panel lines, rivets, roughness variation, grime, MLI foil). Normal maps are "derivative bumps" computed in the shader from the same height functions.

(stored by output name). - **IDENTIFY / IDENTIFY ALL** flashes the role name on its monitor. - **APPLY WINDOW LAYOUT** moves the windows.

FLIGHT MODEL

real physics, friendly assists

NEWTONIAN 6-DOF FLIGHT MODEL (SHIP_SIM.GD)

- n-body point-mass gravity (Earth + Moon), dominant body chosen by sphere of influence
- main engine / booster with pressure-dependent Isp and mass flow
- RCS translation + rotation, reaction wheels, engine gimbal and aero control torques
- exponential atmosphere: drag, lift, sideslip force, static stability, damping
- Sutton-Graves stagnation heating with radiative cooling
- spring/damper ground contacts at real gear/hull points (tipping, friction, impacts)
- Flight-assist modes change how stick input becomes torque; they never fake the physics.

SHIP PARAMETER	VALUE
Dry mass / max fuel	10 t / 9 t
Main engine	280 kN, Isp 300 s (sea level) to 345 s (vacuum)
Booster (launch)	1150 kN, 46 t propellant
RCS	4.5 kN, 450 kg monopropellant
Rotation	reaction wheels 60 kN·m + RCS 45 kN·m; max 22°/s pitch/yaw, 40°/s roll

FLIGHT-ASSIST KNOB (PILOT)

- **STAB** stops the ship from spinning.
- **PROGRADE** points the nose along the NAV vector.
- **HOLD** locks the nose where it is.
- **OFF** is real space: nothing slows you down!

The physics is real. The assists make it easy.

BEFORE EVERY MISSION

1. Turn the MASTER POWER key.
2. Flip the overhead switches **in order, left to right**: BATTERY, FUEL PUMPS, AVIONICS, APU, GUIDANCE, IGNITION ARM. Each one lights green when it is done in the right order (CHECKLIST page).
3. Both crew sit down and say "harness on!". The commander checks, clicks HARNESSES on MISSION CONTROL, then presses START.

CREW TIPS (CREW GUIDE)

- Go slow near things. The scoring loves gentle contact.
- If the MASTER CAUTION flashes, Ops looks at SYSTEMS. The fault list says what to do.
- The console's NAV page shows your orbit. On the Commander's MISSION CONTROL, **WARP** speeds time up while coasting. It drops back to ×1 when an engine fires or you get low. In the air it is limited to ×4.
- Homecoming is the longest mission: about 11 minutes, or about 4½ minutes using WARP ×4 during the high glide.

MISSION 1 · FIRST LIGHT

Launch to orbit

Our satellite needs a ride to space! Power up the Falcon, count down, blast off, and fly into a circle around the Earth. Then open the cargo bay and let the satellite go.

PILOT

Arm and light the engine at T-5, then fly the green marker up and over toward the east. Cut the engine when APOAPSIS reaches 90 km, then burn again at the top to make the orbit round.

Hero moment: Lighting the engine at T−5 and steering into orbit.

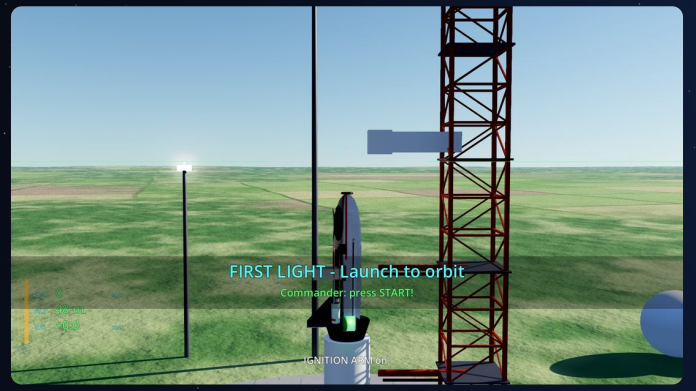
OPS

Watch the booster fuel. When it runs out press your TRIGGER to STAGE. In orbit: open the CARGO DOORS, switch to ARM mode, and press TRIGGER to deploy the satellite.

Hero moment: Staging the booster and releasing the satellite with the arm.

OBJECTIVES, IN ORDER

1. PILOT COUNTDOWN: ENGINE ARM, then IGNITION and full throttle at T-5
2. PILOT Fly the green ascent marker (or PROGRADE assist + NAV GUIDANCE)
3. OPS Booster empty! Press TRIGGER (RCS mode) to STAGE
4. PILOT Burn until APOAPSIS is 90 km, then throttle to zero
5. BOTH Coast up to apoapsis (Commander: time warp on NAV page)
6. PILOT Circularise: burn PROGRADE until PERIAPSIS is above 72 km
7. OPS CARGO DOORS open, mode ARM, TRIGGER to deploy the satellite



On the pad: lattice tower and countdown banner.



Ascent, chase view.



Ascent from the Pilot's window: minimal HUD.

PASS in the flight test: the robot crew completed 7/7 objectives in 229s, 3 stars, hull 100%, max 4.4 g.

MISSION 2 · SPACE HUG

Rendezvous & docking

The space station needs supplies. The Falcon is parked just in front of it. Line up perfectly and glide in slower than a snail to dock.

PILOT

Keep the Falcon pointed straight at the docking port. Use HOLD assist to lock the nose, and fix small drifts with the stick.

Hero moment: Keeping the nose dead-on with HOLD.

OPS

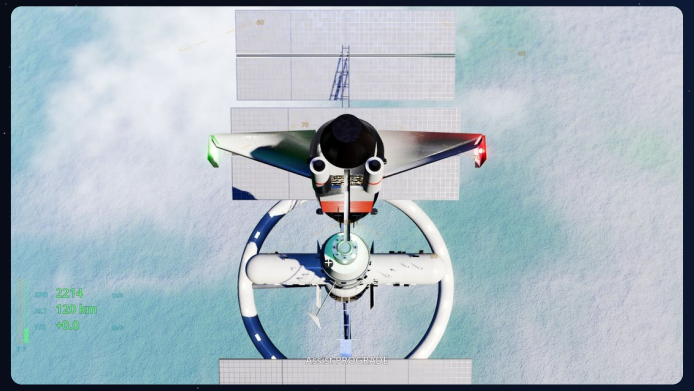
Lock the target, then fly the Falcon sideways, up, down and forward with the RCS stick. Centre the crosshair on the port and creep in. Flip DOCK CLAMP when we touch.

Hero moment: Creeping in with the RCS stick on the docking camera, then DOCK CLAMP!

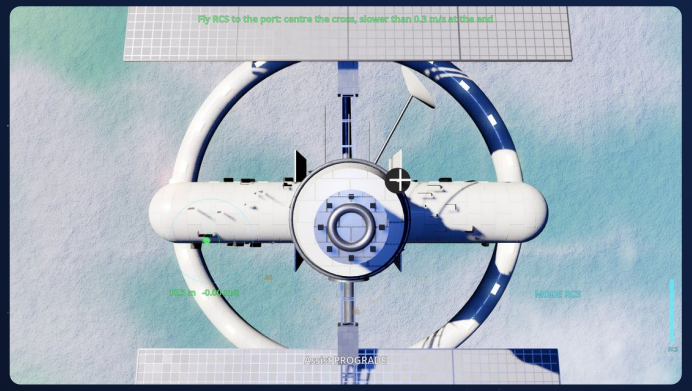
OBJECTIVES, IN ORDER

1. OPS SENSORS on, then NAV TARGET until it says STATION PORT
2. PILOT Line up the nose on the port (HOLD assist helps)
3. OPS Fly RCS to the port: centre the cross, slower than 0.3 m/s at the end
4. OPS Flip DOCK CLAMP to lock on!

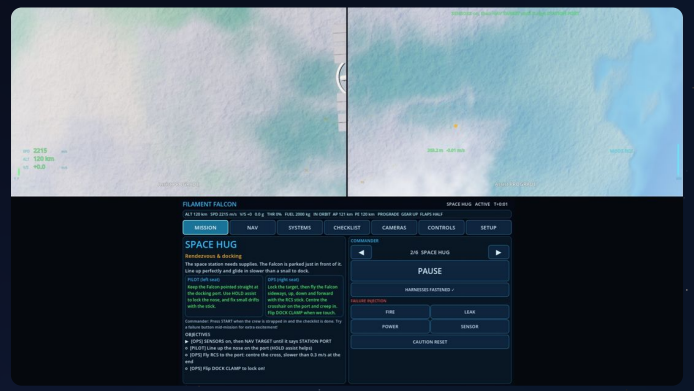
PASS in the flight test: the robot crew completed 4/4 objectives in 189s, 3 stars, hull 100%, max 0.9 g.



Final approach to the station port.



Docking from the window view.



Station in orbit, dev-mode tiled view.

MISSION 3 · EAGLE EYE Lunar landing

We are skimming over the Moon at 9 km. Find the landing beacon, slow down, and land as gently as a feather. No air here, so only the engine can slow us down.

PILOT

Use PROGRADE assist (the NAV vector is set to RETROGRADE) and burn to slow down. Then hover down with the throttle, GEAR DOWN, and touch down slower than 3 m/s.

Hero moment: Slowing down and hovering to a soft touchdown.

OPS

Ping the SCANNER to find the beacon and lock it as NAV TARGET. Near the ground, slide the Falcon over the beacon with the RCS stick.

Hero moment: Finding the beacon with the scanner and sliding the ship over it.

OBJECTIVES, IN ORDER

1. OPS SENSORS on, SCANNER PING to find the landing beacon
2. PILOT Braking burn: PROGRADE assist (RETROGRADE), ENGINE ARM, IGNITION, slow below 60 m/s
3. PILOT Descend: keep the sink rate in the green, below 300 m
4. BOTH GEAR DOWN, hover down gently. Ops: RCS over the beacon

PASS in the flight test: the robot crew completed 4/4 objectives in 281s, 3 stars, hull 100%, max 1.8 g.



Lunar descent from the window.



Hovering down toward the beacon.

MISSION 4 · ROCK HOPPERS

Asteroid rescue & mining

A tiny escape pod is lost in the asteroid field! Find it, fly in carefully, and pull it aboard with the tractor beam. Then mine three glowing crystals and jump home.

PILOT

Fly between the rocks to the pod without crashing (go slow!). Then point at the glowing crystal rocks and hold your TRIGGER to fire the mining laser.

Hero moment: Dodging rocks and lasering crystal rocks.

OPS

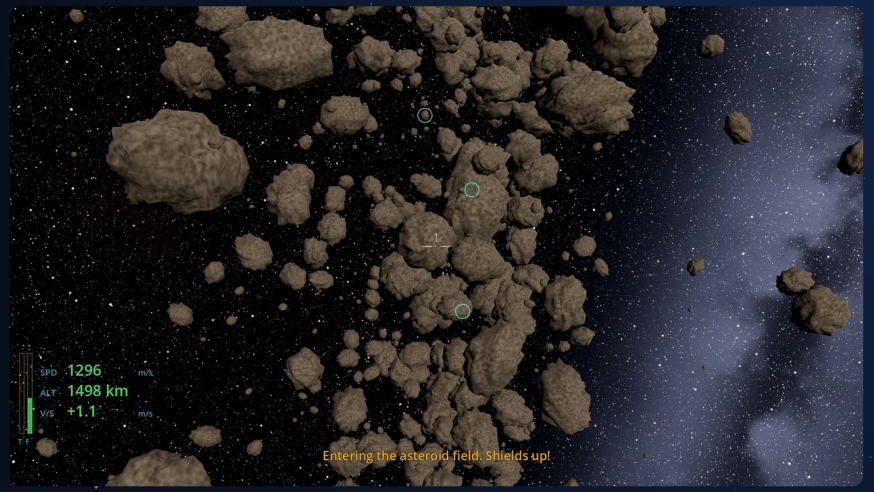
SHIELDS on! Ping the SCANNER to find the pod. Open the CARGO DOORS, choose ARM mode, aim with your stick and HOLD TRIGGER for the tractor beam. Last: arm and engage the JUMP drive.

Hero moment: Tractor-beaming the pod and crystals aboard, then the JUMP!

OBJECTIVES, IN ORDER

1. OPS SHIELDS + SENSORS on, then SCANNER PING to find the pod
2. PILOT Fly to the pod: stop within 40 m (slow near rocks!)
3. OPS CARGO DOORS open, mode ARM, HOLD TRIGGER to beam the pod in
4. BOTH Mine 3 crystals: Pilot lasers glowing rocks, Ops beams crystals in
5. BOTH JUMP ARM, Pilot points at the JUMP marker, Ops ENGAGE!

PASS in the flight test: the robot crew completed 5/5 objectives in 500s, 2 stars, hull 100%, max 0.9 g.



Inside the asteroid field.



Threading between the rocks.

MISSION 5 · HOMECOMING

Reentry & runway landing

Time to come home! We hit the air at over a kilometre per second. The air squeezes and glows like fire around us. Hold the nose up, survive the heat, then glide like a giant paper plane onto the runway.

PILOT

Hold the nose up on the ENTRY bracket (PROGRADE assist with NAV GUIDANCE does it for you). When we slow down, fly through the green boxes to the runway, GEAR DOWN, land, and AIRBRAKE to stop.

Hero moment: Holding the nose up, then flying the green boxes down to the runway.

OPS

SHIELDS on before the heat! After the radio blackout, lock NAV TARGET: RUNWAY and ping the SCANNER to find the landing beacon. Call out the speed and height for the pilot.

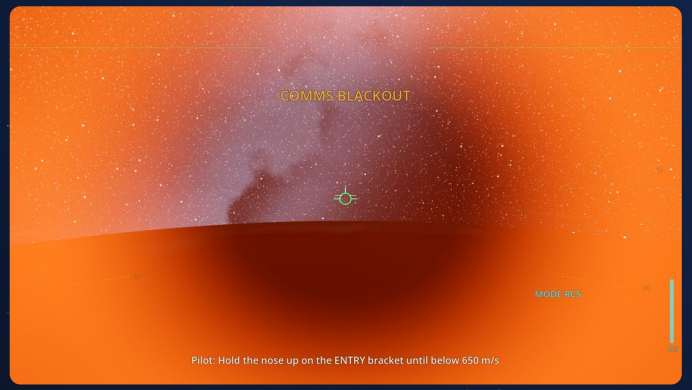
Hero moment: Shields, beating the radio blackout, finding the runway.

OBJECTIVES, IN ORDER

1. OPS SHIELDS on before the heat!
2. PILOT Hold the nose up on the ENTRY bracket until below 650 m/s
3. OPS NAV TARGET: RUNWAY, then SCANNER PING to lock the beacon
4. PILOT Glide through the boxes, GEAR DOWN, touch down on the runway
5. PILOT AIRBRAKE on and roll to a stop



Reentry plasma: SHIELDS!



Plasma from the cockpit.



Runway approach.



Gliding down to the runway.

PASS in the flight test: the robot crew completed 5/5 objectives in 658s, 3 stars, hull 100%, max 2.7 g.

MISSION 6 · FREE FLIGHT Sandbox

No rules, just flying! Visit the station, skim the Moon, explore the asteroid field, glide to the runway, or hop off the launch pad. The jump drive takes you to the next place.

PILOT

Fly anything you like. Try ASSIST OFF to feel real space physics: nothing slows you down!

Hero moment: Try ASSIST OFF!

OPS

Run the systems, play with the arm and scanner, and arm + engage the JUMP drive to travel to the next place.

Hero moment: Play with the arm, scanner and jump drive.

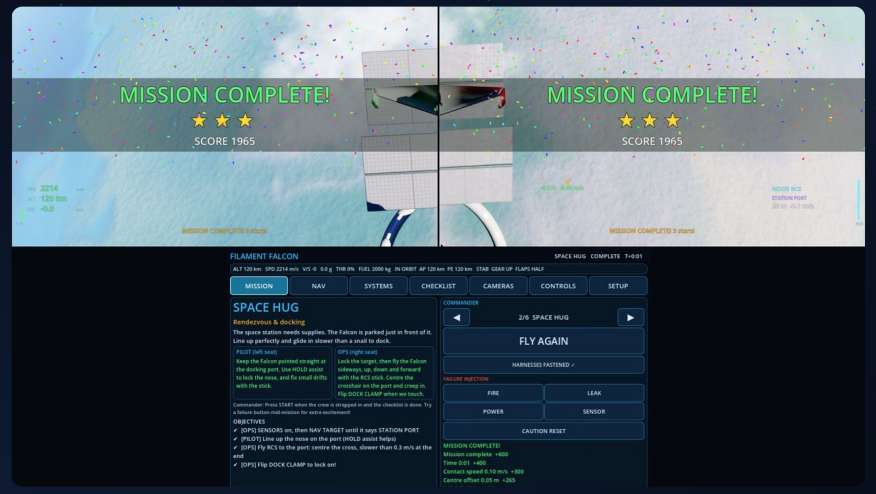
OBJECTIVES, IN ORDER

1. ALL Free flight! Ops: JUMP to visit places. Commander: END FLIGHT on the MISSION page

PASS in the flight test: the robot crew completed 1/1 objectives in 54s, 2 stars, hull 100%, max 1.5 g.



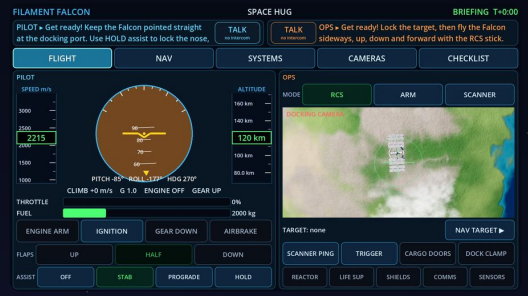
Free flight near the station.



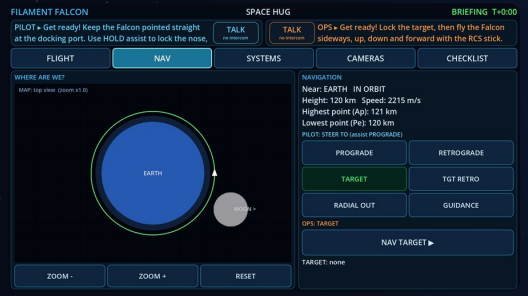
Mission complete: score, stars and celebration.

KIDS' TOUCHSCREEN CONSOLE

ViewSonic TD2220 in the centre console



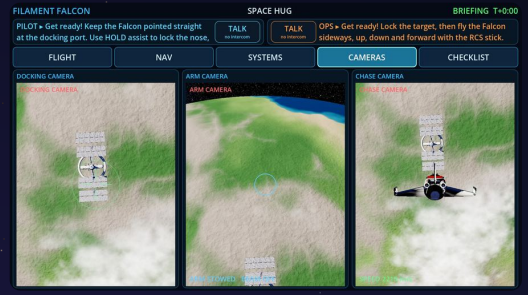
FLIGHT · attitude ball, speed/altitude tapes, fuel; Pilot and Ops controls, live camera



NAV · orbit map, steer-to vectors, nav target



SYSTEMS · power distribution, ship health, the fault list for MASTER CAUTION



CAMERAS · docking, arm and chase camera feeds



CHECKLIST · the overhead start-up switches, green when done in order

MADE FOR 22" TOUCH, ONLY FOR THE KIDS

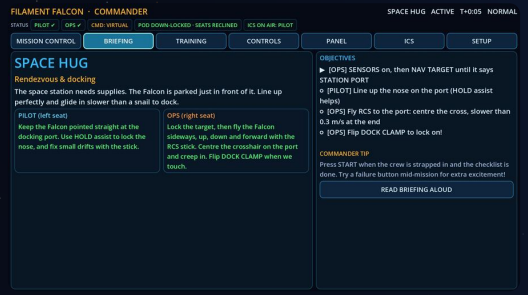
Buttons 76-100 px (19-25 mm), text 22-34 px, readable from the seats. A task strip shows each kid's current job; MASTER CAUTION flashes at the top. When a Falcon board is connected, its physical switches win and the matching touch buttons only mirror them. Touch is mapped to this monitor by `game/tools/setup_displays.sh` (run by `run.sh`).

COMMANDER SCREEN

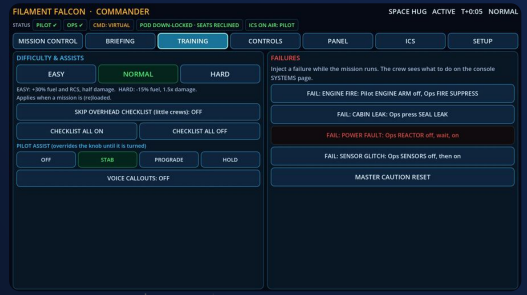
31.5" monitor at dad's seat, mouse and keyboard



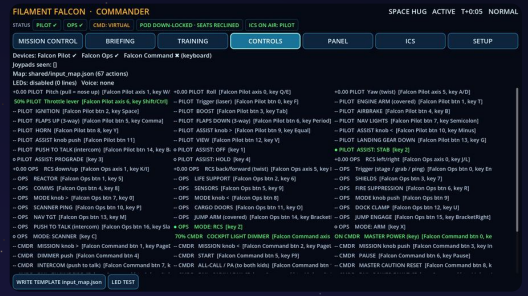
MISSION CONTROL · missions, start/pause/restart, failures, score; orbit map, warp; both kids' tasks, ship health



BRIEFING · briefing, both jobs, every objective, read aloud



TRAINING · difficulty, skip checklist, assist override, voice



CONTROLS · live test of every switch and axis, devices, LED test



SETUP · screen roles, quality, FPS cap, vsync, geometry, LED bridge



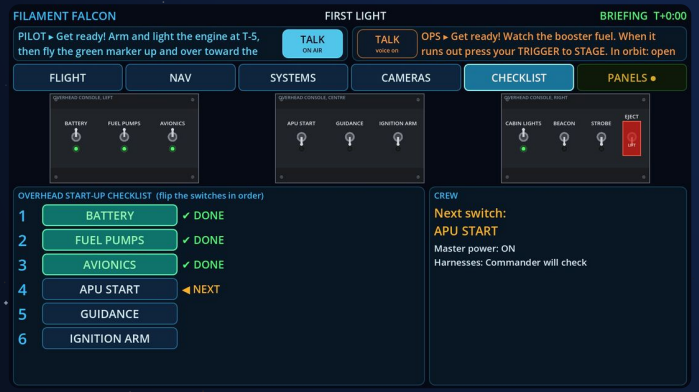
3-SCREEN FALLBACK · no Commander monitor: the same UI as a console tab (Esc)

VIRTUAL PANELS AND INTERCOM

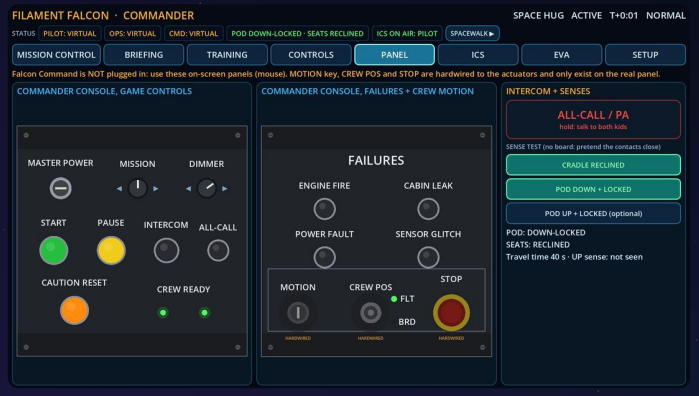
on-screen stand-ins for unplugged boards; three-headset ICS



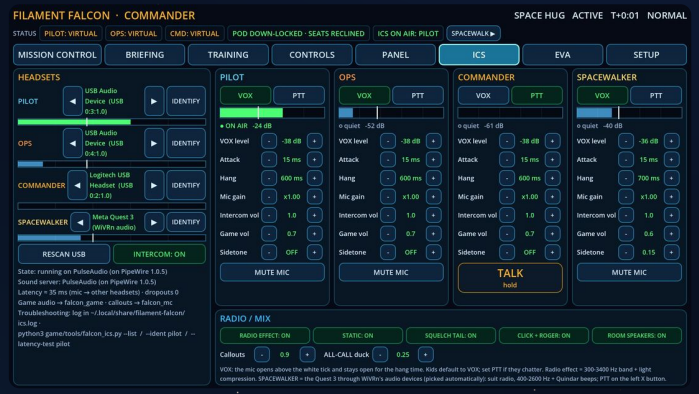
VIRTUAL PANELS · Pilot and Ops boards unplugged: their real panels on the touchscreen, with touch sticks



CHECKLIST + OVERHEAD · Falcon Command unplugged: the three overhead panels on screen



PANEL · CMD_A + CMD_B on screen (mouse); MOTION, CREW POS and STOP stay hardwired-only

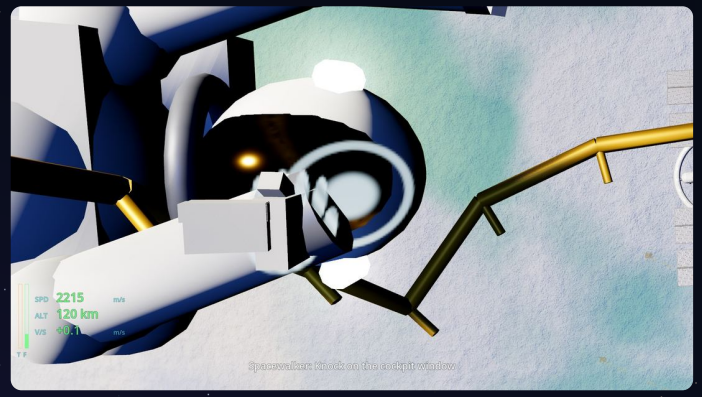


ICS · intercom: headset per seat, VOX/PTT, levels, volumes, radio effect

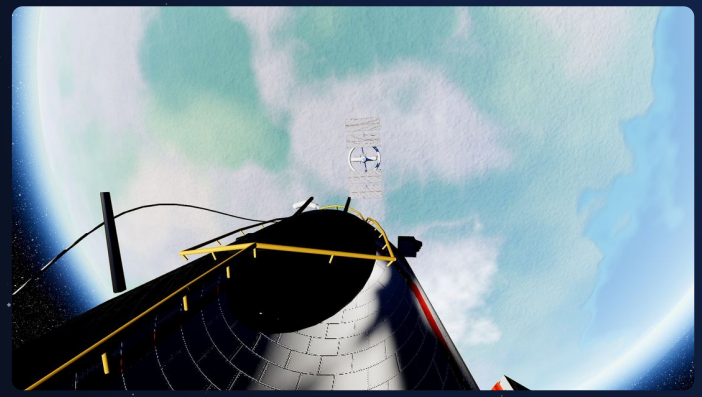
Virtual panels are generated from the CAD layouts (`game/tools/gen_virtual_panels.py`) and appear and disappear as boards are plugged in; guarded switches need two taps (lift the cover, flip). The intercom helper (`game/tools/falcon_ics.py`) mixes three USB headsets on PulseAudio or PipeWire with about 30-40 ms delay; the game's callouts ride on its MISSION CONTROL channel.

VR SPACEWALKS

Meta Quest 3 over USB · WiVRn · a separate VR process



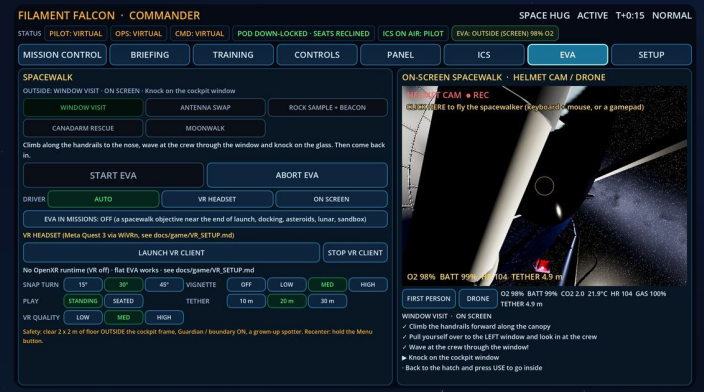
LEFT WINDOW · the spacewalker's glove on the glass, as the Pilot sees it



SPACEWALKER · on the roof rail: nose, handrails, tether, the station and Earth



CONSOLE EVA · HELMET CAM, EVA DRONE, suit telemetry and tasks



COMMANDER EVA · spacewalks, START/ABORT, VR client, comfort options, safety

A fourth player wears a **Meta Quest 3** (USB cable, **WiVRn** on the PC) and goes outside: handrails, a SAFER jetpack, a tether, tools and samples, with Earth, the Moon and the sun around them. The crew sees the spacewalker in real time in the LEFT/RIGHT window views and every camera feed (docking, arm, the EVA drone); the astronaut can wave, point, knock on the cockpit window or put a glove on the glass. Ops can catch a drifting spacewalker with the robotic arm while the Pilot holds attitude.

- **No headset?** Everything works on screen: Commander → **EVA** → driver ON SCREEN → START EVA, then click the helmet cam and fly with the keyboard/mouse or a gamepad. The game never needs an OpenXR runtime; the VR status is on Commander → SETUP and EVA.
- **Architecture:** the headset runs in a **separate process** of the same project (`./run.sh --eva-vr` , or LAUNCH VR CLIENT on Commander → EVA), which streams the astronaut's pose to the cockpit over UDP 47820 at ~90 Hz; the cockpit interpolates it (45 ms buffer). The cockpit itself never loads OpenXR, so its four windows stay smooth and a headset problem cannot crash the flight.
- **Spacewalks:** WINDOW VISIT, ANTENNA SWAP (station), ROCK SAMPLE + BEACON (asteroids), MOONWALK (1/6 g), CANADARM RESCUE (Ops + Pilot + spacewalker). With EVA IN MISSIONS on, missions add one by themselves.
- **Console:** an **EVA** tab appears during a spacewalk: HELMET CAM, EVA DRONE, suit telemetry (O2, battery, CO2, suit temperature, heart rate, jet gas, tether) and the task list.
- **Intercom:** the Quest's speaker and mic (WiVRn's virtual devices) are the 4th seat, **SPACEWALKER**, with a "suit radio" sound (narrower band, more static, Quindar beeps); see section 6.

EVA TESTS (SMOKE TEST, NO HEADSET)

- `XR off in the cockpit process (no runtime needed): ok [No OpenXR runtime (VR off) · flat EVA works · see docs/game/VR_SETUP.md]`
- `EVA window ok steps 5/5 score 900 path 21.0 m (max 5.4 m from hatch) 29 s sim events ["step_done", "step", "knock", "palm", "wave", "eva_done"] in window view: ["left"]`
- `EVA rescue ok steps 3/3 score 900 path 6.0 m (max 7.9 m from hatch) 4 s sim events ["arm_on", "step_done", "step", "arm_off", "eva_done"]`
- `EVA antenna ok steps 6/6 score 1110 path 73.9 m (max 22.1 m from hatch) 109 s sim events ["pick", "step_done", "step", "drop", "tool", "placed", "eva_done"]`
- `EVA VR-client link (headless client, XR unavailable -> autopilot): connected=true xr=false stream 87 Hz, avatar moved 11.3 m, max frame jump 0.209 m, steps 3: ok`

Not testable without the Quest 3: the real OpenXR session, controller bindings, frame timing and the WiVRn audio devices (docs/game/VR_SETUP.md).

TEST RESULTS headless Godot

FLIGHT TEST · ROBOT CREWS FLY EVERY MISSION

	MISSION	OBJECTIVES	SCORE	STARS	HULL	TIME	MAX G
PASS	launch	7/7	1889	★★★	100%	229s	4.4
PASS	docking	4/4	1872	★★★	100%	189s	0.9
PASS	lunar	4/4	1937	★★★	100%	281s	1.8
PASS	asteroids	5/5	1550	★★	100%	500s	0.9
PASS	reentry	5/5	1843	★★★	100%	658s	2.7
PASS	sandbox	1/1	854	★★	100%	54s	1.5

6/6 missions flown to completion. The bots use only the crew's controls (sticks, throttle, switches, buttons, assist modes): no teleporting, real physics.

SMOKE TEST · EVERY MISSION START TO FINISH

	MISSION	OBJECTIVES	SCORE	STARS	HULL
PASS	launch	7/7	3013	★★★	99%
PASS	docking	4/4	3129	★★★	99%
PASS	lunar	4/4	3477	★★★	99%
PASS	asteroids	5/5	2964	★★★	99%
PASS	reentry	5/5	3464	★★★	99%
PASS	sandbox	1/1	1538	★★★	99%

6/6 missions passed, with every injected failure (engine fire, cabin leak, power fault, sensor glitch) fixed by crew actions.

Run 2026-10-11 22:57 ADT (Atlantic time) with Godot 4.7.2.stable.official.ed1daf0bf (headless, Dummy audio) on commit c13e4cb. Re-run with `python3 docs/book/build_book.py --run-tests --godot <path>`, or by hand as in docs/game/README.md section 9.

WHAT IT COSTS AND WHERE TO BUY IT

12 BILL OF MATERIALS

CATEGORY	LINES	TO BUY	TOTAL CAD
Computer & displays (owned)	7	0	$0.00
Video & USB cabling	7	7	$123.69
Controllers & logic	16	16	$295.28
Panel controls	9	9	$210.91
LED backlighting	8	8	$177.14
Printed parts: filament & inserts	5	5	$213.50
Frame lumber & sheet goods	5	5	$676.07
Fasteners & general hardware	10	10	$207.32
Motion system: pivots, shafts, bearings	9	9	$295.28
Motion system: drive & locks	12	12	$519.44
Motion system: hardwired interlock	11	11	$388.84
Motion system: guards	2	2	$92.15
Seats & harnesses	8	8	$248.17
Seats & harnesses - ALTERNATIVE (not in totals)	1	1	$111.98
Grand total to buy (excl. ALTERNATIVE)	**109**	**102**	**$3,447.79**

Recomputed from `bom/Filament_Falcon_BOM.xlsx` exactly like its formulas: Qty to buy = 0 when "Have it" is TRUE, line total = Qty to buy × unit price. Prices before tax and shipping.

$3,447.79

to buy, CAD before tax; with NB HST (15 %) about $3,964.96

BY PRICE SOURCE

PRICE STATUS	CAD
Verified amazon.ca price	$2,027.72
Kent.ca web price (Fredericton store)	$676.07
Estimate - get a quote	$355.00
Estimate - check in store	$236.00
No suitable listing found - estimate	$140.00
Link verified; price NOT shown (placeholder estimate)	$13.00

BY STORE

STORE	CAD
Amazon.ca	$2,180.72
Kent	$676.07
Kent / Home Depot	$236.00
Local metal supplier	$205.00
Industrial supplier	$150.00

USING THE BOM SPREADSHEET

HOW TO USE THE SPREADSHEET

1. Open `bom/Filament_Falcon_BOM.xlsx` in LibreOffice Calc or Excel. Sheets: **BOM**, **Summary**, **Amazon cart**, **Notes**.
2. On **BOM**, set **Have it** to TRUE for anything already in the workshop (switches, wire, screws...). **Qty to buy** drops to 0 and every total updates.
3. **Summary** shows lines, lines still to buy and CAD per category, plus the grand total and how much of it rests on verified prices versus estimates.
4. **Amazon cart** lists only the Amazon.ca rows still to buy (alternatives excluded) with product and search links: work down it to fill the cart.
5. Rows marked "Estimate" are not verified prices: check them in store (Kent, Home Depot) or get a quote (steel, actuators).
6. A CSV copy (`bom/Filament_Falcon_BOM.csv`) and the scripts that built the sheet are in `bom/`.
7. After editing, re-run this book's generator: the BOM chapter and the headline numbers update.

NOTES SHEET

- Sources: filament-falcon/docs/SPEC.md; ff-fw/docs/electronics/README.md; filament-falcon (branch pod-single, frame v0.3 single pod actuator): frame/BOM.md v0.3, frame/cut_list.md, docs/mechanical/README.md, switch_bom.md, displays.md.
- Filament method: python trimesh volume V and surface area A of every 3MF (panels: black panel object + white letters object, including the backing boxes; printed parts x quantities from docs/mechanical/README.md). Printed volume = shell + infill x (V - shell), shell = min(V, A x 1.2 mm) (3 perimeters x 0.4 mm, also used as top/bottom skin proxy); infill 25 % (monitor standoffs 50 %, harness hub 45 %). PETG 1.27 g/cm3, +15 % for purge/brim/failed prints.
- Filament result: black 3601 cm3 -> 5.26 kg (buy 6 x 1 kg); white 44 cm3 -> 0.06 kg (buy 1 x 1 kg). Red/grey parts (guard lids, throttle grip, gear arm, adapter plates) are counted in black - print them in red/grey if you have it. PLA would need ~2 % less mass, but PETG is required for the harness hub and recommended everywhere.
- Prices: "Verified amazon.ca price" = read from the amazon.ca product page on 2026-10-08 (buy-box price, CAD, before tax). Kent prices were read from kent.ca on 2026-10-09 with the Fredericton store selected (Fredericton South, 809 Bishop Dr, and Fredericton North, 25 Two Nations Crossing showed identical prices); they can change, check before buying. Local metal-supplier rows have Google search links for Fredericton NB. Rows marked Estimate/placeholder are NOT verified prices.
- Motion system (v0.3, branch pod-single): 2 cradle actuators 10"/1500 N (~364 mm retracted, self-locking); ONE pod actuator 12"/1500 N self-locking (420 mm retracted, left casing) driving both pod cheeks through a 2x2x1/8" steel torque tube; 2 gas struts ~1050 N with M8 eye ends, sized so the pod is lift-biased (actuator or clevis-pin failure sends it UP, never down); 25 mm shafts, UCP205/UCF205/SHF25; spring pawls + latch plates + Bowden release; steel hard stops; key, E-stop, 20 A fuse, hold-to-run DPDT, relays as 2 H-bridges, one pod limit set (left cheek), roller limit switches, safety edge. Check actuator retracted length and self-locking before ordering.
- SAFETY: the B0B3J1GCVQ buckles are plastic luggage buckles (~50 kg quoted break load, not safety-rated). Positioning harness only for a slow reclining seat. A rated metal-buckle 4-point harness is listed as an ALTERNATIVE row (excluded from totals; switch if you prefer).
- Displays: GPU DP1 -> C275B, HDMI -> C275W (HDMI/VGA only), DP2 -> DP-to-DVI-D -> TD2220 (VGA/DVI only), USB-A -> USB-B for touch. GPU USB-C is spare (4th display max).
- Owned items: PC, both Sceptres, TD2220, 3 HY300, 2 joysticks have Have it = TRUE. Buttons/switches/LEDs you may own are FALSE - tick them.

COULD CHEAP PROJECTORS WRAP THE VIEW AROUND THE CREW?

13 PROJECTOR DOME STUDY



Concept only: three HY300 projectors rear-projecting onto a 1.2 m (3' 11") radius dome band around the pod pivot (`cad/assembly/dome_concept.scad`).

RECOMMENDATION

Do not build the dome now; keep the two Sceptre monitors in the pod. With the drop-down pod the screens are already 540-590 mm in front of the kids' eyes and the pod itself hides ~70 % of any dome behind it, a 3-projector HY300 dome would be ~38 nits (vs ~250 for the monitors), parallax-wrong for two kids sitting 920 mm apart, needs more video outputs than the 6800 XT has left, and makes the room footprint ~4.6 × 4.5 m. If you want the "space around us" feeling later, the cheap, low-risk add-on is **one HY300 front-projecting an ambient starfield / planet onto the ceiling or the wall beyond the cockpit** from the spare USB-C output (one Godot window, no blending, no warp beyond keystone), or a translucent "side window" panel at the left casing. A real dome is only worth it as a v2 that replaces the monitors (option D below) with 3+ brighter 1080p projectors.

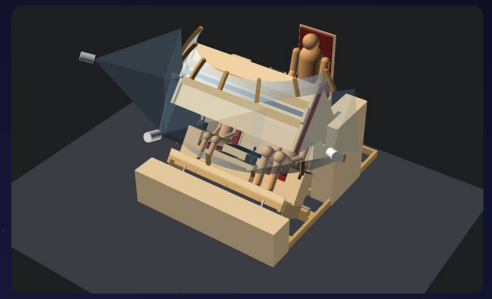
VERDICT

Keep the monitors. If wanted, add option A (one HY300 ambient projection onto the ceiling above/forward of the pod, or a wall) - it uses the spare USB-C output, needs no blending and gives the "you are in space" feeling in the kids' peripheral vision. Revisit a real dome (option D, 3 × 1080p ≥ 1000 lm short-throw projectors) only if the cockpit is rebuilt without the monitor pod.

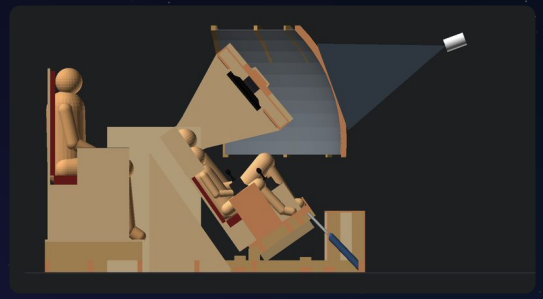
OUTPUTS AND FIT WITH THE COCKPIT

OPTIONS FOR THE SPARE GPU OUTPUT

OPTION	OUTPUTS	VERDICT
A. 1 ambient projector on USB-C → HDMI	4/4	**recommended add-on**: no blending, one extra Godot window
B. 3 projectors via one DP → 3 x HDMI splitter (Matrox TripleHead2Go-class / "video wall" box, 3840 × 720 or 5760 × 1080 surface)	4/4	works, splitter ~150-300 $, Godot renders one wide window and warps it in thirds
C. Second GPU or a second PC (mini PC) running the dome as a networked view client	3 + n	most flexible, most work
D. Dome replaces the monitors: 3 projectors on DP1, DP2/HDMI, TD2220 on USB-C → DVI	4/4	only as v2 with brighter 1080p units; pod becomes a panel frame with an open window



Dome concept, ¾ view.



Side view.

WHY NOT WITH THE DROP-DOWN POD

- The dome band is concentric with the pod pivot, so the pod swing (≤ 712 mm) never reaches it. But its lower edge (~0.9 m above the floor) runs across the side entry gaps in front of the casings (at the seats it is ~1.0 m out from the centre line), so the kids would duck under it to board; the dome ends would have to hinge or start higher (elevation ≥ 10°).
- **Occlusion**: in flight the pod sits 600 mm in front of the kids. From the PILOT's eye it blocks azimuth ~ -32..+65° and elevation ~6..70°; from OPS the mirror image. What is left of a ±75° dome are two side strips (~40° each) and a sliver below the pod (where the legs and the nose box are) → ~70 % of the dome is hidden. That is the main reason not to build it with the drop-pod design.
- Projectors 1 m outside the dome need stands/ceiling mounts out of reach (hot lenses, cables across the floor) and add ~1 m to every side of the footprint.
- Cost (3 HY300 + fabric + ribs + splitter) ~400-600 CAD, plus significant calibration time.

AS OF OCTOBER 12, 2026

14 STATUS AND NEXT STEPS

STATUS (README)

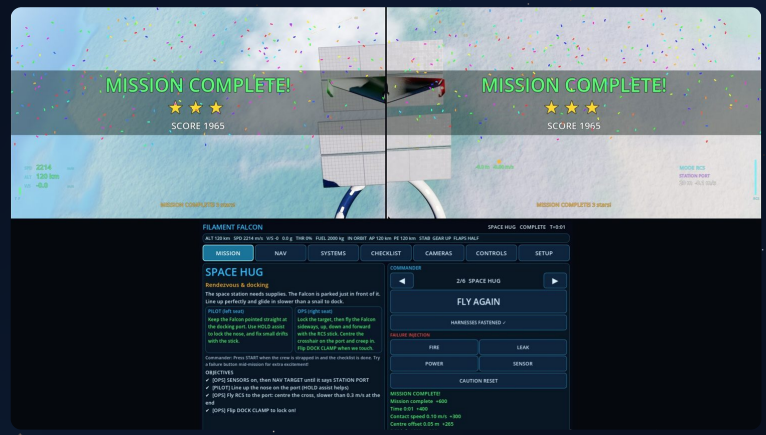
v0.1 design and software. Nothing has been physically printed, wired or flashed yet.

DONE ON PAPER AND IN SOFTWARE

- Master spec; CAD for 11 panels and 31 printed part files, all 3MF exports validated.
- Parametric wooden frame with motion geometry, clearance checks, drawings, cut list and frame BOM (v0.3: reclining cradle + drop-down pod on ONE actuator with a steel torque tube).
- ESP32-S3 firmware for the three boards, wiring diagrams, udev rules, LED protocol; crew-position circuit (relay H-bridges, interlock, one pod limit set) with three read-only position senses.
- Three-headset intercom (VOX / PTT, radio effect, ALL-CALL, MISSION CONTROL callouts) and on-screen virtual panels for unplugged boards.
- VR spacewalks for a Meta Quest 3 (WiVRn, separate VR process streaming the astronaut to the cockpit windows and cameras at ~90 Hz), five EVAs, a 4th intercom seat with a suit radio, and an on-screen EVA mode.
- Godot game v0.1: six missions, four-screen setup (two windows, kids' touch console, Commander screen). Tests: 6/6 missions passed (smoke), 6/6 missions flown to completion (flight).
- Editable BOM with 110 lines: $3,447.79 CAD still to buy.

NOT DONE YET

- Nothing printed, cut, wired or flashed.
- Real LED segment map (measure the strip with `FIND <index>`).
- Real measurements of joysticks, buckles, VESA screw depth and the TD2220 body.
- Actuators, struts and steel parts ordered and checked against the model.
- Game on the real three screens and touch panel (screenshots so far come from the software renderer).
- VR spacewalk with the real Quest 3 (tested so far without a headset).



The goal: a crew that sees this screen after their first real flight.

OPEN QUESTIONS FOR YVES

DISPLAYS (DISPLAYS.MD)

- Which monitor is PILOT (left) vs OPS (right). The renders assume C275B (144 Hz, DP) left, C275W right.
- Actual VESA depth/screw length on both Sceptres; C275W is VESA **75**.
- TD2220 adapter choice (DP → DVI-D passive or USB-C → DVI-D) and cable lengths from the PC location.

ELECTRONICS (ELECTRONICS README)

- "3-way" joysticks are three **potentiometers** (X, Y, twist) + one button. If they turn out to be switch (digital) sticks, wire each direction to a 74HC165 input or spare GPIO instead and map them as buttons in the config header.
- Pot values 5-100 kΩ are all fine at 3.3 V.
- DevKitC-1 v1.0 has its RGB LED on GPIO48 (v1.1: GPIO38, change `PIN_STATUS_RGB`).
- USB IDs 1209:0003/0004/0005 are pid.codes test PIDs: fine for a private build, not for products.
- LED segment map is a placeholder: label ids and order from CAD's `shared/panel_labels.json` (LED count = CAD's estimate at 60/m), 6-LED gaps between panels, and each panel's 5 mm status LEDs right after its labels. Measure the real strip with `FIND <index>` and edit `shared/led_segments.json` (or re-run `firmware/tools/make_placeholder_segments.py`), then rebuild Falcon Command.
- The 5 mm status LED holes in the CAD panels are driven as WS2812D-F5 addressable LEDs on the same data line (no extra firmware outputs). If you prefer plain LEDs, they would need a 74HC595 chain instead.

BEFORE PRINTING (MECHANICAL README)

- Printer bed ≥ 220 × 220 mm, with multi-material (two filaments: black panel, white letters).
- Monitors: Sceptre C275B-144R (VESA 100) and C275W-1920RN (VESA 75), 1800R, see `displays.md`.
- Touchscreen: ViewSonic TD2220, 511.5 × 310.4 × 66.5 mm - measure and set `TD`, `GAP`, `CUT` in `touchscreen_bezel.scad`.
- Harness buckles: 35 × 33 × 13 mm female bodies assumed - measure and set them in `harness_hub.scad`.
- Joystick mounting pattern unknown - measure, then set `SQ_PITCH` / `CENTER_D` in `joystick_housing.scad` (slots already cover bolt circles 44-88 mm).
- Switch bushings: 6 mm mini toggles, 12/16/19/22 mm buttons, EC11 (7 mm), 19 mm key switch. Hole sizes are in `FF_CUTS` in `ff_lib.scad`; hole compensation `FF_HOLE_COMP = 0.2`.
- LED strip 12 mm wide (10 mm fits), 60 LED/m; ~136 LEDs estimated for 67 labels.

NOTICED WHILE COMPILING THIS BOOK

- **Cabling plan vs. the real rig.** docs/mechanical/displays.md plans the C275W (OPS) on HDMI, but on the owner's rig HDMI-A-0 drives the 31.5" Commander monitor and the windows are on DisplayPort-2/-1. The game maps roles by output name and size, so this works. Update the cabling table when the final cables are in.

SUGGESTED BUILD ORDER

based on the frame assembly steps and the open items

01 · DECIDE AND MEASURE

Answer the open questions: which monitor goes left, harness choice (kit buckles vs. rated metal), joystick type and bolt pattern, buckle size, and a ceiling ≥ 1 950 mm (6' 4.8") where the cockpit will live.

02 · ORDER

Work down the BOM's Amazon cart; get quotes for steel plates, shafts and actuators. Check retracted lengths (364 / 415 mm) and self-locking before ordering actuators.

03 · BENCH ELECTRONICS

Flash one DevKitC-1 with falcon_joystick_test, wire diagram 01, calibrate, check it in evtest / jstest-gtk and on the game's CONTROLS page.

04 · PRINT TEST PIECES

One panel (black body, white letters with the backing boxes) for fit and light leaks, one harness hub base with the real buckles, one joystick housing.

05 · BUILD THE FRAME

Base, nose box, towers and casings, cradle on its shaft (frame assembly steps 1-6). Test the recline empty.

06 · POD AND COUNTERBALANCE

Pod cheeks, back sheet and bands; gas struts with the pod up, then the pod actuators (steps 7-9).

07 · INTERLOCK

Wire the hardwired interlock (frame drawing 04); test every limit, lock, key, E-stop, lever and the safety edge with nobody seated (step 10).

08 · GUARDS AND GAPS

Brush strips, skins, crush-zone ramp; re-check every gap with the 25 mm dowel and the 120 mm ball (step 11).

09 · PANELS, DISPLAYS, HARNESS

Bolt in the panels and the monitors with their safety cables, route cables through the pivot clamps, fit the harnesses (step 12).

10 · LEDS AND GAME

Lay the strip, measure the real segment map, rebuild Falcon Command; set up xrandr, autostart the touch mapping and set the display geometry on the Commander SETUP page.

11 · FIRST FLIGHT

Checklist, harnesses on, hands on laps, heads back, hold FLT... and launch First Light.

VERSION AND CHANGELOG

PROJECT HISTORY (GIT LOG)

COMMIT	DATE (ATLANTIC TIME)	CHANGE
5858783	2026-10-11 22:58	Book: VR spacewalks page, EVA test results (tests run on c13e4cb)
c13e4cb	2026-10-11 22:56	VR spacewalks for Meta Quest 3 (WiVRn): separate EVA client, full-body avatar in windows/cams, 5 EVAs, 4th ICS seat, flat mode, docs + tests
56614b2	2026-10-11 22:03	run.sh: re-import after updates (stale class cache after git pull broke the game)
c861bf4	2026-10-10 23:32	Intercom buttons on the CAD panels, 3 position senses, BOM + book
89e93c3	2026-10-10 23:26	Merge branch 'pod-single'
b52ef3d	2026-10-10 23:26	ICS intercom (3 USB headsets, VOX/PTT, radio FX, ALL-CALL, mission-control channel), virtual panels + sticks for unplugged boards, crew-position v0.2 (pod/cradle senses, callouts, diagram 04), docs + book
0080259	2026-10-10 23:15	Pod: single 1500 N actuator + steel torque tube, lift-biased struts (frame v0.3)
dac4d8c	2026-10-10 20:27	Game: four-screen cockpit - kids' touch console + Commander screen, auto touch mapping
fdc56f3	2026-10-10 19:51	Rebuild project book
0f811ec	2026-10-10 19:51	game: run.sh launcher imports on first run (fixes parse errors on fresh clone)
48c7c30	2026-10-09 19:08	Location: Fredericton, NB; rebuild BOM and project book
4c7c5a9	2026-10-09 18:30	Add project book PDF and generator
224848a	2026-10-08 22:57	Add BOM spreadsheet and project README
2993d99	2026-10-08 22:57	Merge branch 'game'

THIS EDITION

Book version	1.0
Generated	2026-10-12 09:44 ADT (Atlantic time)
Source	github.com/whyvas/filament-falcon, main @ 5858783 + uncommitted changes (25 commits)
Generator	`docs/book/build_book.py`, `book.css`, `template.html`

BOOK CHANGELOG

- **1.0** (2026-10-09): first edition, generated from the repo after the mechanical v0.2 and game v0.1 merges.

NOT FOUND IN THE REPO AT BUILD TIME

- Nothing: every file, section and image the generator looks for was found.

Regenerate after any change: `python3 docs/book/build_book.py`. Prices in CAD before tax. Metric first, imperial in brackets.